HSC Mathematics Extension 1 & 2

Functions Practice

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Download PDF and resources:
`https://github.com/vuhung16au/math-olympiad-ml`

A function is a rule that assigns to each element of one set exactly one element of another set.

—Mathematical Folklore

Contents

1 Introduction

1.1 Project Overview

This booklet is a growing collection of functions problems for NSW HSC Mathematics Extension 1 students, with a few short Extension 2 preview notes where they help build mathematical maturity. It covers the major themes of the syllabus: understanding what a function is, working with inverses, composing functions, and analysing their graphs. The problems are designed to build fluency with algebraic manipulation and geometric reasoning side by side.

1.2 Target Audience

This resource is written for students who want extra practice with functions in a clear, classroom-friendly format. Solutions are intentionally concise, but each one highlights the key idea—whether that is a domain restriction, a symmetry argument, or a composition rule. The main focus is Extension 1, while some comments point ahead to Extension 2 or university mathematics.

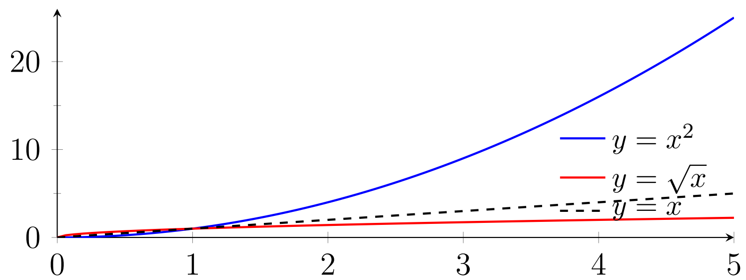
1.3 How to Use This Booklet

- Read the short summary of the core definitions and ideas first.
- Attempt each problem before reading the solution.
- Pay attention to domains and ranges: they are often where mistakes happen.
- Look for symmetry: the graph of an inverse is the reflection of the original in the line $y = x$.
- Use the takeaways to build pattern recognition for future HSC questions.

1.4 Extension 1 Knowledge Needed

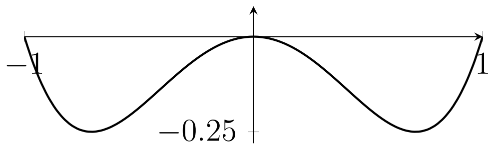
Most problems in this booklet are accessible with standard HSC Mathematics Extension 1 knowledge. The key background ideas are:

- **Function notation:** writing $f(x)$, understanding that f is the rule and $f(x)$ is the value, and reading expressions such as $f(a + h)$ or $f(f(x))$.
- **Domain and range:** identifying the largest possible domain (the natural domain) and the corresponding range, and applying domain restrictions when stated.
- **One-to-one functions:** recognising that a function has an inverse if and only if it passes the horizontal line test.
- **Inverse functions:** swapping x and y to find f^{-1}, and understanding that $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$. For $x \geq 0$, restricting $f(x) = x^2$ gives inverse $f^{-1}(x) = \sqrt{x}$, illustrated below on $[0, 5]$ with $y = x$ (reflection line for the two graphs):

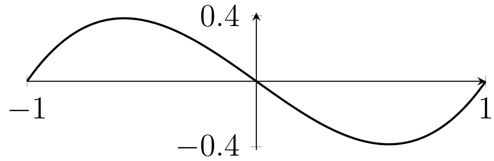


- **Graphs of inverse functions:** reflecting the graph of f in the line $y = x$ to obtain f^{-1}.
- **Composite functions:** forming $f(g(x))$ by substituting one function into another, and identifying the correct domain of the composition.
- **Even and odd functions:** f is even if $f(-x) = f(x)$ for all x (the graph has y-axis symmetry); f is odd if $f(-x) = -f(x)$ for all x (the graph has rotational symmetry about the origin). For example, on $[-1, 1]$, $y = x^4 - x^2$ is even and $y = x^3 - x$ is odd:

$y = x^4 - x^2$ (even)



$y = x^3 - x$ (odd)



- **Transformations:** translations, reflections, dilations, and their effect on the equation and graph of a function.

1.5 Core Language of Functions

Function. A function $f : A \to B$ assigns to every element $x \in A$ exactly one element $f(x) \in B$. The set A is the **domain**, the set B is the **codomain**, and the **range** is the set of outputs that actually occur:

$$\text{range}(f) = \{f(x) : x \in A\} \subseteq B.$$

The empty function. If the domain is the empty set $\varnothing$, then there is exactly one function

$$f : \varnothing \to B,$$

called the **empty function**. It has no inputs and therefore no outputs. Its range is also empty. This is mainly a set-theoretic idea, but it is good to know that it is still a perfectly valid function.

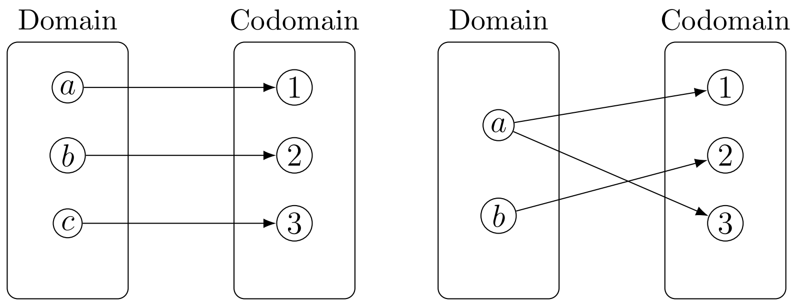
Variable. A variable is a symbol, such as x or t, that stands for a value that may change. In $y = f(x)$, the variable x is the input and the variable y is the output.

Map or mapping. A map is another name for a function or relation that pairs elements of one set with elements of another set.

1.6 Relations and Mappings

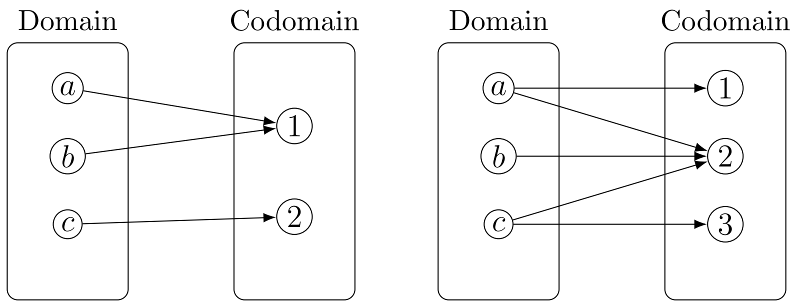
A **relation** between sets A and B is any collection of ordered pairs (a, b) with $a \in A$ and $b \in B$. A function is a special kind of relation in which each input has exactly one output.

The most common mapping types are shown below.



One-to-one
(injective)

One-to-many
(not a function)



Many-to-one
(a function)

Many-to-many
(not a function)

One-to-one and many-to-one relations can be functions. One-to-many and many-to-many relations are not functions because at least one input has more than one output.

1.7 Domain, Range, and Basic Tests

Domain. The domain is the set of allowed input values.

Range. The range is the set of actual output values.

Vertical line test. A curve in the plane is the graph of a function if and only if every vertical line meets the curve at most once.

Horizontal line test. A function is one-to-one (injective) if and only if every horizontal line meets its graph at most once. Only one-to-one functions have an inverse that is also a function.

1.8 Important Number Sets

We often write:

$$\mathbb{N} = \{1, 2, 3, \dots\}, \qquad \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\},$$
$$\mathbb{Q} = \left\{\frac{a}{b} : a, b \in \mathbb{Z},\ b \neq 0\right\}, \qquad \mathbb{R} = \text{the set of real numbers.}$$

Good to know for later: the set of complex numbers is written $\mathbb{C}$, not $\mathbb{Z}$. Complex numbers are studied in Extension 2, so in this booklet they appear only as short preview notes.

1.9 Important Families of Functions

Linear function. A linear function has the form

$$f(x) = mx + c,$$

where m is the gradient and c is the y-intercept. Its graph is a straight line.

Quadratic function. A quadratic function has the form

$$f(x) = ax^2 + bx + c, \qquad a \neq 0.$$

Its graph is a parabola.

Polynomial. A polynomial over $\mathbb{R}$ is an expression of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where the coefficients $a_0, a_1, \ldots, a_n$ are real numbers. If the coefficients are allowed to be complex numbers, then P is a polynomial over $\mathbb{C}$. This is a good-to-know Extension 2 idea.

Cubic function. A cubic function has the form

$$f(x) = ax^3 + bx^2 + cx + d, \qquad a \neq 0.$$

Reciprocal function. A basic reciprocal function is

$$f(x) = \frac{1}{x},$$

which is undefined at $x = 0$.

1.10 Continuity and Differentiation Preview

Continuity. Informally, a function is continuous if its graph can be drawn without lifting your pen. More precisely, f is continuous at $x = a$ if small changes in x produce small changes in $f(x)$.

Continuous function. A function is continuous on an interval if it is continuous at every point of that interval.

Extension 2 preview. In more formal Extension 2 language, f is continuous at $x = a$ if

$$\lim_{x \to a} f(x) = f(a).$$

Derivative. The derivative $f'(x)$ measures the instantaneous rate of change of f and the gradient of the tangent to the graph.

Differentiation. Differentiation is the process of finding the derivative. This is Extension 2 material, but it is good to know because many important functions are later studied through their derivatives.

Sequence of numbers. A sequence is an ordered list of numbers, usually written

$$a_1, a_2, a_3, \ldots$$

or as (a_n), where n is a positive integer.

Limit of a sequence. We say (a_n) has limit L if the terms get closer and closer to L as n becomes very large. We write

$$\lim_{n \to \infty} a_n = L.$$

Why this matters for functions. Sequences help us understand what a function does when x approaches a value or becomes very large. They are one of the basic ideas behind limits, continuity, and later calculus. This is mainly Extension 2 and university material, but it is very useful to know early.

1.11 Symmetry, Inverses, and Composition

Even and odd functions.

$$f \text{ is even} \iff f(-x) = f(x) \text{ for all } x. \qquad f \text{ is odd} \iff f(-x) = -f(x) \text{ for all } x.$$

An even function has symmetry about the y-axis. An odd function has rotational symmetry about the origin.

Inverse function. If f is one-to-one with domain A and range B, then its inverse $f^{-1} : B \to A$ satisfies

$$f^{-1}(f(x)) = x \quad \text{for all } x \in A, \qquad f(f^{-1}(y)) = y \quad \text{for all } y \in B.$$

To find f^{-1}: write $y = f(x)$, swap x and y, then solve for y.

Reflection principle. The graph of f^{-1} is the reflection of the graph of f in the line $y = x$.

Composite function. Given $f : B \to C$ and $g : A \to B$, the composite $f \circ g : A \to C$ is defined by

$$(f \circ g)(x) = f(g(x)).$$

Note that in general $f \circ g \neq g \circ f$.

Domain of a composite. For $f(g(x))$ to be defined, x must be in the domain of g *and* $g(x)$ must be in the domain of f:

$$\mathrm{dom}(f \circ g) = \{x \in \mathrm{dom}(g) : g(x) \in \mathrm{dom}(f)\}.$$

Fixed point. A fixed point of a function f is a value a such that

$$f(a) = a.$$

Graphically, fixed points are the points where the graph of $y = f(x)$ meets the line $y = x$.

1.12 Key Transformations

Given a base function $y = f(x)$, the standard transformations are:

- $y = f(x) + c$: shift up by c (down if $c < 0$).
- $y = f(x - h)$: shift right by h (left if $h < 0$).
- $y = af(x)$: vertical dilation by factor $|a|$; reflection in x-axis if $a < 0$.
- $y = f(bx)$: horizontal dilation by factor $\frac{1}{|b|}$; reflection in y-axis if $b < 0$.
- Replacing y by $-f(x)$ reflects the graph in the x-axis.
- Replacing x by $-x$ gives $y = f(-x)$, which reflects the graph in the y-axis.
- Rotations can also be studied, but they are less central in HSC Extension 1 function questions because a rotated graph need not remain a function.

These transformations combine: $y = af(bx - h) + c$ is the general form.

1.13 Piecewise-Defined Functions

A piecewise function assigns a different rule to different parts of the domain. For example:

$$f(x) = \begin{cases} x^2 & x < 0, \\ 2x + 1 & x \geq 0. \end{cases}$$

Key skills are: evaluating at a given x, finding the range piece by piece, and drawing the graph with correct open/closed endpoints.

1.14 Absolute Value as a Piecewise Function

$$|x| = \begin{cases} x & x \geq 0, \\ -x & x < 0. \end{cases}$$

The graph of $y = |f(x)|$ reflects any portion of $y = f(x)$ that lies *below* the x-axis up above it. The graph of $y = f(|x|)$ keeps the right half of $y = f(x)$ and reflects it to create an even function.

Absolute value of a real number. For a real number x, the absolute value $|x|$ is its distance from 0 on the number line.

Absolute value of a complex number. Good to know for Extension 2 and beyond: if $z = a + bi$, then

$$|z| = \sqrt{a^2 + b^2}.$$

This is the distance of z from the origin in the complex plane. Complex numbers belong to Extension 2, not Extension 1, so this booklet only mentions them briefly.

1.15 Reciprocals, Asymptotes, and Sign Tables

Reciprocal of a function. The reciprocal of $f(x)$ is

$$\frac{1}{f(x)},$$

provided that $f(x) \neq 0$.

Asymptotes. An asymptote is a line that the graph approaches. For example, $y = \frac{1}{x}$ has vertical asymptote $x = 0$ and horizontal asymptote $y = 0$.

Table of signs. A table of signs records where an expression is positive, zero, or negative. It is especially useful for rational functions, inequalities, and sketching graphs.

1.16 Parametric and Multivariable Functions

Parametric definition. Sometimes a relation is defined by a third variable, called a parameter. For example,

$$x = t, \qquad y = \frac{1}{t}, \qquad t \neq 0.$$

This defines the curve $y = \frac{1}{x}$ in parametric form.

Multiple-variable functions. A function can also take more than one input, such as

$$f(x, y) = x^2 + y^2.$$

These are important in higher mathematics but are outside the main Extension 1 and Extension 2 syllabuses.

1.17 Important HSC Functions

The most important functions to recognise quickly in HSC mathematics are:

- linear functions $mx + c$,
- quadratic functions $ax^2 + bx + c$,
- cubic functions,
- reciprocal functions such as $\frac{1}{x}$ and more general rational functions,
- absolute value functions,
- exponential functions such as a^x with $a > 0$ and $a \neq 1$,
- logarithmic functions such as $\log_a x$,
- trigonometric functions such as $\sin x$, $\cos x$, and $\tan x$.

In Extension 1, the main emphasis is on algebraic, exponential, logarithmic, and trigonometric functions. In Extension 2, ideas such as continuity, differentiation, and later complex-number-based viewpoints become more important.

1.18 A Brief Note on Big O Notation

Big O notation is a way of comparing how fast functions grow. It is widely used in university mathematics and computer science.

If we write

$$f(x) = O(g(x)) \quad \text{as } x \to \infty,$$

we mean that for sufficiently large x, the function $f(x)$ grows no faster than a constant multiple of $g(x)$.

This topic is **outside the HSC Mathematics Extension 1 and Extension 2 syllabuses**, but it is very good to know for later university studies.

Examples.

- If $f(x) = 3x^2 + 5x + 7$, then $f(x) = O(x^2)$ because the x^2 term dominates for large x.
- If $f(x) = x^5 - 2x^3 + 1$, then $f(x) = O(x^5)$.
- If $f(x) = \log x$, then $f(x) = O(\log x)$.
- If $f(x) = 2^x + 100x^4$, then $f(x) = O(2^x)$ because the exponential term eventually dominates the polynomial term.

Growth comparison to remember. As $x \to \infty$,

$$\log x \text{ grows more slowly than any positive power of } x,$$

$$x^n \text{ grows more slowly than } a^x \quad \text{for any fixed } n \in \mathbb{N} \text{ and } a > 1.$$

So in rough order of growth:

$$\log x \quad < \quad x,\ x^2,\ x^3, \dots \quad < \quad 2^x,\ 3^x, \dots$$

Polynomial example. If

$$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$$

with $a_n \neq 0$, then

$$P(x) = O(x^n).$$

So the degree of the polynomial tells you its dominant growth rate.

1.19 Problem-Solving Habits for This Booklet

- Always state the domain and range of your answer, not just the rule.
- When finding an inverse, restrict the domain of f first if the function is not one-to-one.
- When composing, work from the inside out: compute $g(x)$ first, then apply f.
- Sketch the graph whenever the question involves domains, ranges, inverses, or asymptotes.
- In even/odd questions, substitute $-x$ and simplify fully before drawing a conclusion.
- For rational expressions, locate any zeroes and undefined points before making a sign table.

1.20 A Small Extension 2 Preview

This booklet is mainly about HSC Mathematics Extension 1. When complex numbers, formal limits, or derivatives are mentioned, they should be read as **good to know** ideas that will appear properly in Extension 2 or in later university mathematics.

Continuity in formal language. For Extension 2 students, a function is **continuous** at $x = a$ if

$$\lim_{x \to a} f(x) = f(a).$$

Piecewise functions need this checked at the break-points. Many Extension 2 questions later ask you to verify continuity or use derivatives to study turning points and behaviour.

Remark 1.1

The most common functions mistake is forgetting to adjust the domain when restricting a function to make it one-to-one before finding its inverse. Always check that your inverse has the correct domain and range.

Remark 1.2 (Notation warning)

$f^{-1}(x)$ means the inverse *function*, not $\dfrac{1}{f(x)}$. If you mean the reciprocal, write $[f(x)]^{-1}$ or $\dfrac{1}{f(x)}$ explicitly.

2 Part 1: Worked Problems

2.1 Basic

Problem 2.1: Rearranging a Relation into a Function

Write the equation

$$1 + 2xy = 0$$

as a function with dependent variable y.

Hint: Solve the equation for y by isolating the term containing y first.

Solution 2.1

Starting from

$$1 + 2xy = 0,$$

we get

$$2xy = -1.$$

If $x \neq 0$, divide both sides by $2x$:

$$y = -\frac{1}{2x}.$$

So the relation can be written as the function

$$\boxed{y = -\frac{1}{2x}}, \qquad x \neq 0.$$

Takeaways 2.1

- To write a relation as a function of x, solve explicitly for y.
- Always state any domain restriction created by division. Here $x \neq 0$.

Problem 2.2: A Difference Quotient

If

$$f(x) = x^2 + 2x,$$

express

$$\frac{f(x+h) - f(x)}{h}$$

in simplest form.

Hint: Expand $f(x+h)$ carefully, then subtract $f(x)$ before dividing by h.

Solution 2.2

$$f(x+h) = (x+h)^2 + 2(x+h) = x^2 + 2xh + h^2 + 2x + 2h.$$

So

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{(x^2 + 2xh + h^2 + 2x + 2h) - (x^2 + 2x)}{h} \\ &= \frac{2xh + h^2 + 2h}{h} \\ &= 2x + h + 2.\end{aligned}$$

Hence

$$\boxed{\frac{f(x+h) - f(x)}{h} = 2x + h + 2}$$

for $h \neq 0$.

Takeaways 2.2

- Subtracting $f(x)$ often causes many terms to cancel.
- The quotient $\dfrac{f(x+h)-f(x)}{h}$ is the starting point of differentiation in Extension 2.

Problem 2.3: A Table and a Quadratic Graph

Given the function

$$y = x^2 - 3x + 2,$$

(i) fill the table for $x = -2, -1, 0, 1, 2$,

(ii) draw the graph of y on an xy-diagram for $-3 \le x \le 3$.

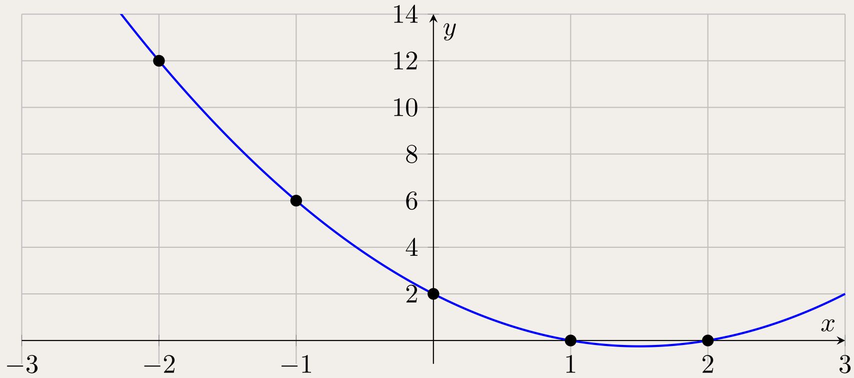
Hint: Substitute each x-value directly into the formula. For the graph, plot the points and join them smoothly to form a parabola.

Solution 2.3

(i) Substituting each value of x:

x	-2	-1	0	1	2
$y = x^2 - 3x + 2$	12	6	2	0	0

(ii) The graph is an upward-opening parabola. It crosses the x-axis at $(1,0)$ and $(2,0)$ and crosses the y-axis at $(0,2)$.

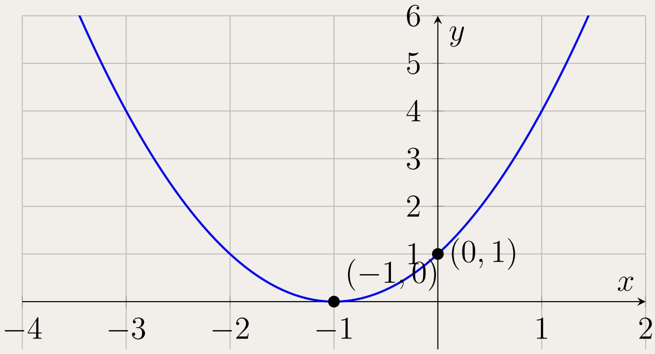


Takeaways 2.3

- A table of values is often the fastest way to begin sketching a graph.
- Quadratic graphs are parabolas, and intercepts are especially important points to plot.

Problem 2.4: Which Quadratic Matches the Graph?

The graph below is a parabola with vertex $(-1, 0)$, touching the x-axis at $(-1, 0)$ and crossing the y-axis at $(0, 1)$.



Which equation matches the graph?

A. $y = (x - 1)^2$

B. $y = x^2 + 1$

C. $y = (x + 1)^2$

D. $y = -(x + 1)^2$

Hint: Read the vertex first. Then check whether the parabola opens upward or downward.

Solution 2.4

The graph has vertex $(-1, 0)$, so the equation must be of the form

$$y = a(x + 1)^2.$$

Since the graph opens upward, $a > 0$. Also the y-intercept is $(0, 1)$, which matches

$$y = (0 + 1)^2 = 1.$$

Therefore the correct choice is

$$\boxed{\textbf{C. } y = (x + 1)^2.}$$

Takeaways 2.4

- Vertex form $y = a(x - h)^2 + k$ reveals the vertex (h, k) immediately.
- A parabola touching the x-axis at its vertex has a repeated root.

Problem 2.5: Four Familiar Functions

Given

$$f_1(x) = \frac{1}{x}, \qquad f_2(x) = x, \qquad f_3(x) = x^2, \qquad f_4(x) = x^3,$$

(i) fill the table for $x = -3, -2, -1, 0, 1, 2, 3,$

(ii) draw the graphs of f_1, f_2, f_3, f_4 on the same xy-diagram, using different colours.

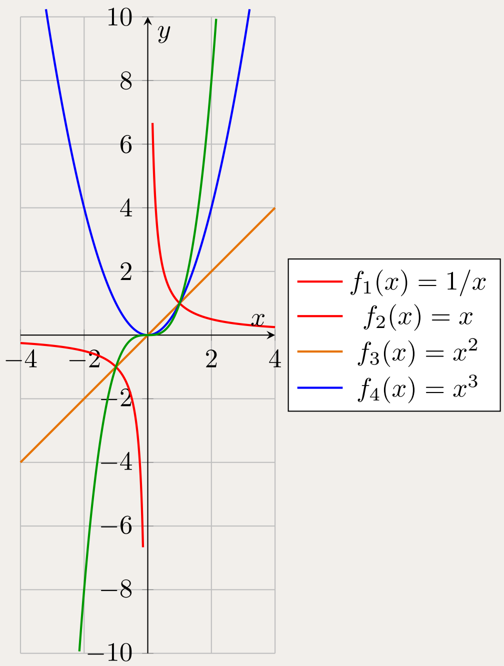
Hint: Remember that $\frac{1}{x}$ is undefined at $x = 0$. Compare the symmetries of the four functions as you graph them.

Solution 2.5

(i) The table is

x	-3	-2	-1	0	1	2	3
$f_1(x) = \frac{1}{x}$	$-\frac{1}{3}$	$-\frac{1}{2}$	-1	undefined	1	$\frac{1}{2}$	$\frac{1}{3}$
$f_2(x) = x$	-3	-2	-1	0	1	2	3
$f_3(x) = x^2$	9	4	1	0	1	4	9
$f_4(x) = x^3$	-27	-8	-1	0	1	8	27

(ii) A combined sketch is shown below.



Takeaways 2.5

- These four graphs are among the most important basic shapes in HSC mathematics.
- x and x^3 are odd functions; x^2 is even; $\frac{1}{x}$ is odd with asymptotes.

Problem 2.6: Zeroes and a Sign Table for a Cubic

Given

$$y = (x+2)(x-1)(x-3),$$

(i) how many zeroes does it have? List them.

(ii) Fill the sign table for y.

(iii) Discuss the behaviour of the function as $x \to -\infty$ and $x \to +\infty$, and sketch the graph roughly.

Hint: Each linear factor gives one zero. For the sign table, test one value in each interval determined by the zeroes.

Solution 2.6

(i) The zeroes occur when one factor is zero:

$$x = -2, \qquad x = 1, \qquad x = 3.$$

So the function has **three** zeroes.

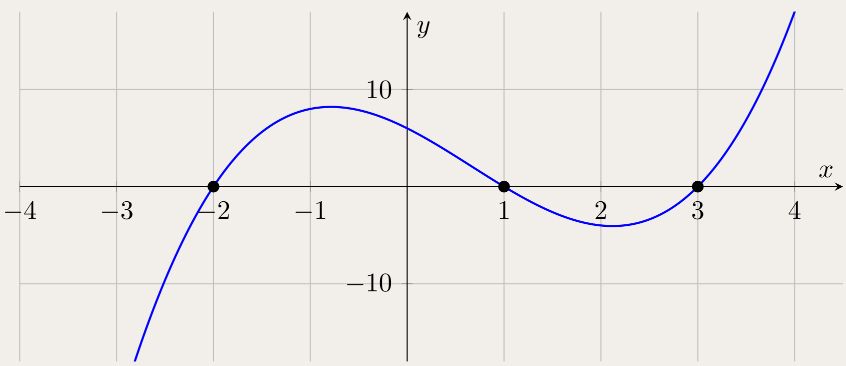
(ii) The critical intervals are $(-\infty, -2)$, $(-2, 1)$, $(1, 3)$, and $(3, \infty)$.

x	$(-\infty,-2)$	-2	$(-2,1)$	1	$(1,3)$	3	$(3,\infty)$
y	$-$	0	$+$	0	$-$	0	$+$

(iii) Since the leading term is x^3 with positive coefficient,

$$y \to -\infty \text{ as } x \to -\infty, \qquad y \to +\infty \text{ as } x \to +\infty.$$

The graph crosses the x-axis at $x = -2, 1, 3$ and alternates sign across these simple roots.

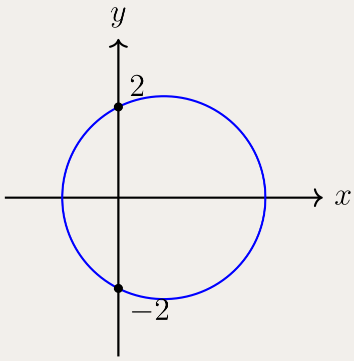


Takeaways 2.6

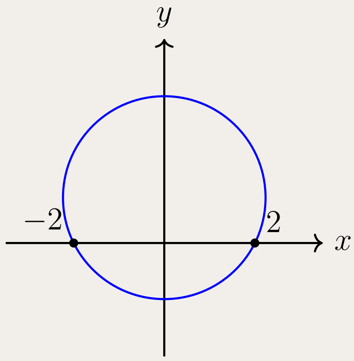
- A factored form reveals zeroes immediately.
- A sign table helps you understand where a function is above or below the x-axis.
- For odd-degree polynomials with positive leading coefficient, the graph falls left and rises right.

Problem 2.7: A Circle Graph Matching Question

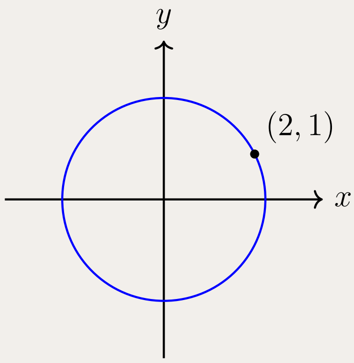
Question: Four graphs of circles are given below, labelled 1 to 4. Four multiple-choice options present pairings of a specific circle graph with an equation. Only one of these pairings is mathematically correct. Which option shows the correct graph-and-equation pair?



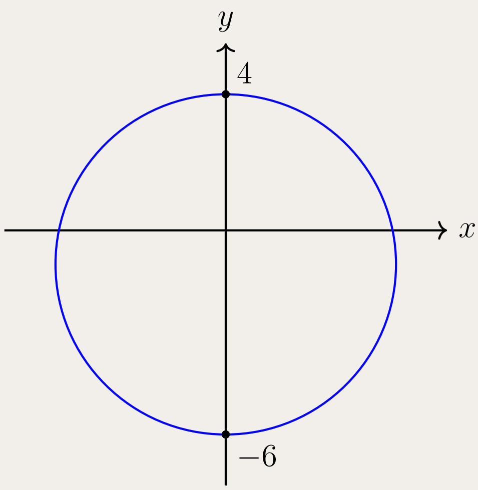
Graph 1



Graph 2



Graph 3



Graph 4

A. Graph 1 is represented by $x^2 + (y-1)^2 = 5$

B. Graph 2 is represented by $x^2 + (y-1)^2 = 5$

C. Graph 3 is represented by $x^2 + (y-1)^2 = 5$

D. Graph 4 is represented by $x^2 + (y+1)^2 = 5$

Hint: For a circle, identify its centre and radius first. The equation

$$(x-a)^2 + (y-b)^2 = r^2$$

has centre (a, b) and radius r.

Solution 2.7

The equation

$$x^2 + (y-1)^2 = 5$$

represents a circle with centre $(0, 1)$ and radius $\sqrt{5}$.
Graph 2 is centred at $(0, 1)$ and passes through $(2, 0)$ and $(-2, 0)$. Since

$$(2-0)^2 + (0-1)^2 = 4 + 1 = 5,$$

its radius is indeed $\sqrt{5}$.
Therefore the correct option is

$$\boxed{\textbf{B.}}$$

Takeaways 2.7

- The equation $(x-a)^2 + (y-b)^2 = r^2$ is the standard form of a circle.
- Matching graphs to equations becomes much easier once you identify centre and radius separately.

2.2 Medium

Problem 2.8: The Nested Quadratic Chain

Consider the quadratic function

$$f(x) = x^2 - 6x + 7.$$

Find all real values of x satisfying

$$f(f(x)) = -1.$$

Hint: Do not expand $f(f(x))$ into a quartic straight away.

- Let $u = f(x)$ and solve $f(u) = -1$ first.
- Then set $f(x) = u$ for each value of u you find.
- Completing the square is a clean way to solve the resulting quadratics.

Solution 2.8

Let $u = f(x)$. Then $f(f(x)) = -1$ becomes

$$f(u) = -1,$$

so

$$u^2 - 6u + 7 = -1 \implies u^2 - 6u + 8 = 0 \implies (u-4)(u-2) = 0.$$

Hence

$$u = 4 \quad \text{or} \quad u = 2.$$

So we solve the two equations:

Case 1: $f(x) = 4$

$$x^2 - 6x + 7 = 4 \implies x^2 - 6x + 3 = 0.$$

Thus

$$x = \frac{6 \pm \sqrt{36-12}}{2} = 3 \pm \sqrt{6}.$$

Case 2: $f(x) = 2$

$$x^2 - 6x + 7 = 2 \implies x^2 - 6x + 5 = 0 \implies (x-1)(x-5) = 0.$$

So

$$x = 1 \quad \text{or} \quad x = 5.$$

Therefore, the real solutions are

$$\boxed{x = 1,\ 5,\ 3-\sqrt{6},\ 3+\sqrt{6}.}$$

Takeaways 2.8

- **Level-by-level solving:** Although $f(f(x))$ would expand to a quartic, it is much better to treat it as a quadratic in disguise by introducing a substitution.
- **Composition structure:** For a composite equation $f(g(x)) = c$, solve the outer layer first, then feed those answers back into the inner layer.
- **Root count intuition:** A quadratic composed with another quadratic can produce up to four real solutions, which is exactly what happens here.
- **Higher-ed connection:** Problems involving $f(f(x))$, $f(f(f(x)))$, and beyond are the first step toward iterated functions, dynamical systems, and fixed-point ideas.

Problem 2.9: Deriving Functional Rules Through Substitution

Let f be a function such that

$$f : \mathbb{R} \setminus \{1\} \to \mathbb{R}.$$

The function satisfies

$$f\left(\frac{x+1}{x-1}\right) = \frac{x}{x+1}$$

for all real numbers $x \neq \pm 1$.

(i) Find the value of x such that $\dfrac{x+1}{x-1} = 2$. Hence calculate $f(2)$.

(ii) By using the substitution $u = \dfrac{x+1}{x-1}$, express $f(x)$ in terms of x wherever the rule is determined.

Hint: For (i), solve the inner equation first, then substitute that value of x into the right-hand side. For (ii), solve

$$u = \frac{x+1}{x-1}$$

for x in terms of u, then substitute into $\frac{x}{x+1}$.

Solution 2.9

(i) Solve

$$\frac{x+1}{x-1} = 2.$$

Then

$$x + 1 = 2x - 2 \implies x = 3.$$

Substitute $x = 3$ into the original equation:

$$f(2) = \frac{3}{3+1} = \boxed{\frac{3}{4}}.$$

(ii) Let

$$u = \frac{x+1}{x-1}.$$

Solving for x,

$$u(x-1) = x + 1 \implies x(u-1) = u + 1 \implies x = \frac{u+1}{u-1}.$$

Hence

$$f(u) = \frac{\frac{u+1}{u-1}}{\frac{u+1}{u-1} + 1} = \frac{\frac{u+1}{u-1}}{\frac{2u}{u-1}} = \frac{u+1}{2u}.$$

Replacing u by x gives

$$\boxed{f(x) = \frac{x+1}{2x}}$$

for $x \neq 0, 1$.
The value $f(0)$ is not determined by the given equation, because the input 0 would require

$$\frac{x+1}{x-1} = 0 \implies x = -1,$$

but $x = -1$ is excluded.

Takeaways 2.9

- The input to f is the whole expression inside the brackets, not the parameter x itself.
- To recover a hidden function rule, invert the inner expression and substitute back into the output side.
- Always check which inputs are actually covered by the substitution; here the equation determines $f(x)$ for $x \neq 0, 1$, but not $f(0)$.

Problem 2.10: Composites with a Parameter

Given the functions

$$f(x) = x^2 - k \qquad (k \in \mathbb{R}), \qquad g(x) = 2x + 1,$$

(a) find simplified algebraic expressions for $f(g(x))$ and $g(f(x))$,

(b) find the exact value of k such that the graphs of $y = f(g(x))$ and $y = g(f(x))$ intersect at exactly one point,

(c) for this value of k, find the coordinates of the intersection point.

Hint: First compute the two composites. Then set them equal and use the discriminant condition for a quadratic to have exactly one real solution.

Solution 2.10

(a) Since $g(x) = 2x + 1$,

$$f(g(x)) = (2x+1)^2 - k = 4x^2 + 4x + 1 - k.$$

Also

$$g(f(x)) = 2(x^2 - k) + 1 = 2x^2 - 2k + 1.$$

(b) Intersection points satisfy

$$f(g(x)) = g(f(x)).$$

So

$$4x^2 + 4x + 1 - k = 2x^2 - 2k + 1$$

which simplifies to

$$2x^2 + 4x + k = 0.$$

For exactly one point of intersection, this quadratic must have exactly one real root:

$$\Delta = 4^2 - 4(2)(k) = 16 - 8k = 0.$$

Hence

$$\boxed{k = 2.}$$

(c) Substituting $k = 2$,

$$2x^2 + 4x + 2 = 0 \quad \Longrightarrow \quad x^2 + 2x + 1 = 0 \quad \Longrightarrow \quad (x+1)^2 = 0,$$

so $x = -1$.
Then

$$y = f(g(-1)) = f(-1) = (-1)^2 - 2 = -1.$$

Therefore the intersection point is

$$\boxed{(-1, -1).}$$

Takeaways 2.10

- Composite functions are often compared by setting their outputs equal.
- “Exactly one intersection” usually leads to a discriminant equal to 0.

Problem 2.11: A Relation That Splits into Two Functions

Consider the relation

$$y^2 - 4xy + 5x^2 = 9.$$

(a) By treating the equation as a quadratic in y, use the quadratic formula to express y in terms of x.

(b) Hence explain why this relation does not represent a single function, but rather two separate function branches.

(c) Determine the maximum domain of x for which this relation is defined.

(d) Find the exact points where the two branches meet.

Hint: Think of the equation as

$$y^2 + (-4x)y + (5x^2 - 9) = 0.$$

The two branches meet when the square-root part becomes zero.

Solution 2.11

(a) Viewing the equation as a quadratic in y,

$$y^2 - 4xy + (5x^2 - 9) = 0.$$

So

$$y = \frac{4x \pm \sqrt{(-4x)^2 - 4(1)(5x^2 - 9)}}{2}.$$

This gives

$$y = \frac{4x \pm \sqrt{16x^2 - 20x^2 + 36}}{2} = \frac{4x \pm \sqrt{36 - 4x^2}}{2} = 2x \pm \sqrt{9 - x^2}.$$

(b) Since

$$y = 2x + \sqrt{9 - x^2} \qquad \text{or} \qquad y = 2x - \sqrt{9 - x^2},$$

the relation gives two possible y-values for most allowed x-values. Therefore it is not a single function, but two branches:

$$f_1(x) = 2x + \sqrt{9 - x^2}, \qquad f_2(x) = 2x - \sqrt{9 - x^2}.$$

(c) For the square root to be defined,

$$9 - x^2 \ge 0 \quad \Longrightarrow \quad x^2 \le 9 \quad \Longrightarrow \quad -3 \le x \le 3.$$

So the maximum domain is

$$\boxed{[-3, 3].}$$

(d) The branches meet when

$$\sqrt{9 - x^2} = 0,$$

so

$$x = \pm 3.$$

Then $y = 2x$, giving the points

$$\boxed{(-3, -6) \quad \text{and} \quad (3, 6).}$$

Takeaways 2.11

- Some algebraic relations split naturally into two branches after solving for y.
- A square root creates a domain restriction and often reveals where branches meet.

Problem 2.12: A Family of Lines Through an Intersection Point

Consider two distinct, non-parallel lines

$$L_1 : a_1x + b_1y + c_1 = 0, \qquad L_2 : a_2x + b_2y + c_2 = 0,$$

with unique intersection point $P(x_0, y_0)$.

(a) Prove that for any real constant k, the equation $L_1 + kL_2 = 0$ also passes through P.

(b) Explain algebraically why $L_1 + kL_2 = 0$ can represent every line through P, except for the line L_2 itself.

(c) The lines $3x - 4y + 2 = 0$ and $x + 2y - 4 = 0$ intersect at a point P. Without finding the coordinates of P, find the equation of the line that passes through both P and the origin.

Hint: At the intersection point P, both L_1 and L_2 equal 0. In part (c), force the family line to pass through $(0, 0)$.

Solution 2.12

(a) Since P lies on both lines,

$$L_1(x_0, y_0) = 0 \qquad \text{and} \qquad L_2(x_0, y_0) = 0.$$

Therefore

$$(L_1 + kL_2)(x_0, y_0) = L_1(x_0, y_0) + kL_2(x_0, y_0) = 0 + k \cdot 0 = 0.$$

So $L_1 + kL_2 = 0$ passes through P for every real k.

(b) Any line through P in this family can be written more generally as

$$\lambda L_1 + \mu L_2 = 0,$$

where λ and μ are not both zero. If $\lambda \neq 0$, divide through by λ to obtain

$$L_1 + \left(\frac{\mu}{\lambda}\right) L_2 = 0.$$

Thus every such line can be written in the form $L_1 + kL_2 = 0$, except the special case $\lambda = 0$, which gives precisely the line

$$L_2 = 0.$$

(c) Write the required line as

$$(3x - 4y + 2) + k(x + 2y - 4) = 0.$$

Since it passes through the origin,

$$2 + k(-4) = 0 \implies 2 - 4k = 0 \implies k = \frac{1}{2}.$$

Hence

$$(3x - 4y + 2) + \frac{1}{2}(x + 2y - 4) = 0.$$

Multiplying by 2 gives

$$6x - 8y + 4 + x + 2y - 4 = 0,$$

so the line is

$$\boxed{7x - 6y = 0.}$$

Takeaways 2.12

- The linear combination $L_1 + kL_2 = 0$ is a standard way to describe a whole pencil of lines through a common point.
- This method often avoids solving for the actual intersection point.

Problem 2.13: Why a Quadratic Cannot Have Three Distinct Roots

Show that the quadratic equation

$$ax^2 + bx + c = 0 \qquad (a \neq 0)$$

cannot have more than two distinct roots.

Hint provided in the question: suppose the equation has three distinct roots α, β, γ, then substitute them into the equation and derive a contradiction. Also explain the result visually using the graph of a quadratic function.

Hint: Write down the three equations obtained from α, β, γ. Subtract pairs of equations to eliminate c.

Solution 2.13

Suppose, for contradiction, that

$$ax^2 + bx + c = 0$$

has three distinct roots α, β, γ.
Then

$$a\alpha^2 + b\alpha + c = 0, \qquad a\beta^2 + b\beta + c = 0, \qquad a\gamma^2 + b\gamma + c = 0.$$

Subtract the first two equations:

$$a(\alpha^2 - \beta^2) + b(\alpha - \beta) = 0.$$

Since $\alpha \neq \beta$,

$$(\alpha - \beta)\big(a(\alpha + \beta) + b\big) = 0 \quad \Longrightarrow \quad a(\alpha + \beta) + b = 0.$$

Similarly,

$$a(\beta + \gamma) + b = 0.$$

Subtracting these gives

$$a(\alpha - \gamma) = 0.$$

But $\alpha \neq \gamma$, so this forces

$$a = 0,$$

which contradicts the assumption $a \neq 0$.
Therefore a quadratic cannot have three distinct roots.
Visual explanation. The graph of $y = ax^2 + bx + c$ is a parabola. A parabola can intersect the x-axis at at most two points, so the equation $ax^2 + bx + c = 0$ can have at most two real roots.

Takeaways 2.13

- A degree-2 polynomial has at most two distinct roots.
- The algebra matches the geometry: a parabola meets the x-axis at most twice.

Problem 2.14: Symmetry About the Vertex

Consider the quadratic function

$$f(x) = x^2 - 2x - 8.$$

(a) By completing the square, express $f(x)$ in vertex form and state the coordinates of the vertex.

(b) Let $x = v$ be the x-coordinate of the vertex. Simplify $f(v + h)$ and $f(v - h)$.

(c) What does the result prove about the geometric shape of the parabola relative to its vertex?

Hint: Complete the square first. Then use the vertex x-coordinate v as the centre of the two shifted inputs.

Solution 2.14

(a)

$$f(x) = x^2 - 2x - 8 = (x-1)^2 - 9.$$

So the vertex is

$$\boxed{(1, -9).}$$

(b) Here $v = 1$. Then

$$f(1+h) = (1+h)^2 - 2(1+h) - 8 = h^2 - 9,$$

and

$$f(1-h) = (1-h)^2 - 2(1-h) - 8 = h^2 - 9.$$

Hence

$$\boxed{f(v+h) = f(v-h).}$$

(c) Since points equally spaced to the left and right of $x = v$ give the same y-value, the parabola is symmetric about the vertical line

$$\boxed{x = v = x_{\text{vertex}}.}$$

So the axis of symmetry is $x = 1$.

Takeaways 2.14

- Completing the square reveals the vertex and axis of symmetry immediately.
- The identity $f(v+h) = f(v-h)$ is a precise algebraic expression of mirror symmetry for a quadratic.
- **The Cubic Connection:** A strikingly similar observation is seen with cubic functions. While a quadratic has line symmetry perfectly balanced around its vertex, a cubic polynomial has point symmetry (rotational symmetry) balanced perfectly around its point of inflection. If μ is the x-coordinate of a cubic's inflection point, stepping h units left and right reveals the identity $f(\mu+h) + f(\mu-h) = 2f(\mu)$. The algebraic logic of exploring symmetry through $+h$ and $-h$ shifts remains exactly the same.

Problem 2.15: A Family of Quadratics and the Locus of the Vertex

Consider the family

$$y = x^2 - 2kx + (k^2 + k - 2),$$

where k is a real parameter.

(a) Express the function in vertex form.

(b) Hence write down the vertex in terms of k.

(c) Find the Cartesian equation of the locus of the vertex as k varies, and describe it geometrically.

(d) Find the x-intercepts in terms of k, and state the restriction on k for which they exist.

(e) Find the exact range of k for which the parabola has one strictly positive and one strictly negative x-intercept.

Hint: Complete the square in x. For part (e), think about what the signs of the two roots tell you about their product.

Solution 2.15

(a)

$$y = x^2 - 2kx + k^2 + k - 2 = (x-k)^2 + k - 2.$$

(b) Therefore the vertex is

$$\boxed{(k,\ k-2).}$$

(c) Let the vertex be (x, y). Since $x = k$, we have

$$y = k - 2 = x - 2.$$

So the locus is

$$\boxed{y = x - 2,}$$

which is a straight line.
(d) Put $y = 0$:

$$x^2 - 2kx + (k^2 + k - 2) = 0.$$

Using the completed-square form,

$$(x-k)^2 = 2 - k.$$

Hence

$$x = k \pm \sqrt{2-k}.$$

These are real only when

$$2 - k \geq 0 \quad \Longrightarrow \quad \boxed{k \leq 2.}$$

(e) For one root to be positive and the other negative, their product must be negative:

$$k^2 + k - 2 < 0.$$

Factorising,

$$(k+2)(k-1) < 0.$$

Therefore

$$\boxed{-2 < k < 1.}$$

Takeaways 2.15

- A parameter family often becomes much clearer after completing the square.
- Tracking the vertex is a standard route to finding a locus.
- Opposite-sign roots correspond to a negative product.

Problem 2.16: Composition of One-to-One Functions

Prove or disprove the statement:

The composition of two one-to-one functions is always another one-to-one function.

Hint: To prove a function is one-to-one, start from

$$(f \circ g)(x_1) = (f \circ g)(x_2)$$

and use the fact that both f and g are one-to-one.

Solution 2.16

The statement is **true**.
Let f and g be one-to-one functions. We want to show that

$$f \circ g$$

is also one-to-one.
Assume

$$(f \circ g)(x_1) = (f \circ g)(x_2).$$

Then

$$f(g(x_1)) = f(g(x_2)).$$

Since f is one-to-one, equal outputs imply equal inputs. Therefore

$$g(x_1) = g(x_2).$$

Now use the fact that g is one-to-one. Again, equal outputs imply equal inputs, so

$$x_1 = x_2.$$

Hence

$$(f \circ g)(x_1) = (f \circ g)(x_2) \implies x_1 = x_2,$$

which proves that $f \circ g$ is one-to-one.
Therefore, the composition of two one-to-one functions is always one-to-one.

Takeaways 2.16

- To prove a composition is one-to-one, peel the functions off from the outside in.
- Injective functions preserve distinctions: different inputs cannot collapse to the same output.
- This result helps explain why inverse functions behave well under composition.

Problem 2.17: Domain Overlaps and One-to-One Testing

Consider the following two relations, defined for all real numbers x:

$$S(x) = \begin{cases} 2x & \text{for } x \text{ rational,} \\ x^3 & \text{for } x \text{ irrational,} \end{cases}$$

$$T(x) = \begin{cases} x & \text{for } x^2 \text{ rational,} \\ -x & \text{for } x \text{ irrational.} \end{cases}$$

(a) Determine whether $S(x)$ and $T(x)$ are valid functions. Give reasons.

(b) For any relation from part (a) that is a valid function, decide whether it is one-to-one or many-to-one.

(c) For any valid function, find all values of x such that $f(x) = x$.

Hint: For a valid function, every input must produce exactly one output. In $T(x)$, check whether the two conditions can overlap. In part (b), try to find two different inputs for $S(x)$ that both produce the same output.

Solution 2.17

(a) Since every real number is either rational or irrational, but not both, the two cases of $S(x)$ are disjoint and cover all real numbers. Hence

$$\boxed{S(x) \text{ is a valid function.}}$$

For $T(x)$, take $x = \sqrt{2}$. Then x is irrational, so one rule gives

$$T(\sqrt{2}) = -\sqrt{2}.$$

But $x^2 = 2$ is rational, so the other rule gives

$$T(\sqrt{2}) = \sqrt{2}.$$

One input gives two outputs, so

$$\boxed{T(x) \text{ is not a valid function.}}$$

(b) To classify $S(x)$, note that

$$S(1) = 2$$

and, since $\sqrt[3]{2}$ is irrational,

$$S(\sqrt[3]{2}) = (\sqrt[3]{2})^3 = 2.$$

Thus

$$S(1) = S(\sqrt[3]{2}) \quad \text{with} \quad 1 \neq \sqrt[3]{2},$$

so $S(x)$ is not one-to-one. Therefore

$$\boxed{S(x) \text{ is many-to-one.}}$$

(c) Solve $S(x) = x$ branch by branch.
If x is rational, then

$$2x = x \implies x = 0,$$

and this is valid since 0 is rational.
If x is irrational, then

$$x^3 = x \implies x(x^2 - 1) = 0 \implies x = 0, \pm 1.$$

But these are all rational, so none is allowed in the irrational branch.
Hence the only fixed point is

$$\boxed{x = 0.}$$

Takeaways 2.17

- In a piecewise definition, the cases must give exactly one output for each input.
- It is not enough for the rules to look different; you must check whether their conditions overlap.
- When solving within a branch of a piecewise function, always test whether the solution really belongs to that branch.

2.3 Advanced

Problem 2.18: The Exponential Functional Equation

Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x) > 0$ for all real numbers x and

$$f(x+y) = f(x)f(y) \qquad \text{for all } x, y \in \mathbb{R}.$$

(i) Show that $f(0) = 1$ and deduce that $f(-x) = \dfrac{1}{f(x)}$.

(ii) By using mathematical induction, show that $f(nx) = [f(x)]^n$ for all $n \in \mathbb{Z}^+$.

(iii) Prove that $f(rx) = [f(x)]^r$ for all rational numbers r.

(iv) Suppose that f is continuous on $\mathbb{R}$. Prove that $f(tx) = [f(x)]^t$ for all real numbers t. Hence deduce that $f(x) = a^x$ for some constant a.

Hint: This is a variation of Cauchy's functional equation and is the standard route to exponential functions.

- For (i), substitute $y = 0$, then choose $y = -x$.
- For (ii), write $f((k+1)x)$ as $f(kx + x)$.
- For (iii), let $r = \dfrac{p}{q}$ and compare $f(px)$ with $f\left(q \cdot \dfrac{p}{q}x\right)$.
- For (iv), approximate a real number t by rationals $r_n \to t$ and use continuity to pass to the limit.

Solution 2.18

(i) From $f(x) = f(x+0) = f(x)f(0)$ and $f(x) > 0$, we get $f(0) = 1$. Then

$$1 = f(0) = f(x + (-x)) = f(x)f(-x),$$

so $f(-x) = \dfrac{1}{f(x)}$.

(ii) Induct on $n \in \mathbb{Z}^+$. The case $n = 1$ is trivial. If $f(kx) = [f(x)]^k$, then

$$f((k+1)x) = f(kx + x) = f(kx)f(x) = [f(x)]^{k+1}.$$

Hence $f(nx) = [f(x)]^n$ for all $n \in \mathbb{Z}^+$.

(iii) Using (i) and (ii), the formula extends to all integers:

$$f(0x) = f(0) = 1 = [f(x)]^0, \qquad f(-nx) = \frac{1}{f(nx)} = [f(x)]^{-n}.$$

Now let $r = \dfrac{p}{q}$ with $p \in \mathbb{Z}$ and $q \in \mathbb{Z}^+$. Then

$$[f(x)]^p = f(px) = f\left(q \cdot \frac{p}{q}x\right) = \left[f\left(\frac{p}{q}x\right)\right]^q.$$

Since $f > 0$, taking the positive qth root gives

$$f\left(\frac{p}{q}x\right) = [f(x)]^{p/q}.$$

Thus $f(rx) = [f(x)]^r$ for all $r \in \mathbb{Q}$.

(iv) Let $t \in \mathbb{R}$ and choose rationals $r_n \to t$. By continuity of f,

$$f(tx) = \lim_{n\to\infty} f(r_n x).$$

Using (iii),

$$f(r_n x) = [f(x)]^{r_n}.$$

Since $f(x) > 0$, the map $u \mapsto [f(x)]^u$ is continuous, so

$$f(tx) = \lim_{n\to\infty} [f(x)]^{r_n} = [f(x)]^t.$$

Now put $x = 1$. If we let $a = f(1) > 0$, then

$$f(t) = f(t \cdot 1) = [f(1)]^t = a^t.$$

Hence $f(x) = a^x$ for some constant $a > 0$.

Takeaways 2.18

- **Foundation of exponentials:** Once continuity is added, the functional equation $f(a+b) = f(a)f(b)$ forces the function to be exponential. In particular, if $f(1) = c$, then part (iv) shows that $f(t) = c^t$ for every real number t.
- **Variation of Cauchy's functional equation:** This is the multiplicative analogue of the classical additive equation. A common linear version says that if a continuous function $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x) + f(y) = f(x+y)$ for all real numbers x and y, then $f(x) = ax$ for some constant a.
- **Density plus continuity:** The key step is to prove the rule first for integers, then rationals, and finally extend it from $\mathbb{Q}$ to all real numbers by using the density of the rationals together with continuity.
- **Difficulty note:** This is an olympiad-style problem. You may find a variation of it in a strong trial exam from a top school, but it is unlikely to appear in an official HSC Mathematics Extension 1 or Extension 2 examination in this full form.

Problem 2.19: The Harmonic Conjugates of the Golden Segment

The interval PQ has length p. A point A lies internally between P and Q such that

$$PQ \cdot QA = PA^2.$$

A second point B lies externally on the extension of PQ beyond Q, and A and B are harmonic conjugates with respect to P and Q, meaning

$$\frac{PA}{AQ} = \frac{PB}{BQ}.$$

(i) Let $PA = x$. Show that

$$x = \frac{p(\sqrt{5}-1)}{2}.$$

(ii) Let $PB = y$. Use the harmonic-conjugate ratio to find y in terms of p.

(iii) Hence prove that

$$AB = 2PQ.$$

Hint: Write AQ and BQ in terms of x, y, and p. In part (ii), substitute the value of x from part (i) into the ratio equation.

Solution 2.19

(i) Let $PA = x$. Then $AQ = p - x$, so

$$p(p-x) = x^2 \implies x^2 + px - p^2 = 0.$$

Thus

$$x = \frac{-p \pm p\sqrt{5}}{2}.$$

Since $x > 0$,

$$\boxed{x = \frac{p(\sqrt{5}-1)}{2}.}$$

(ii) Let $PB = y$. Then $BQ = y - p$, and

$$\frac{x}{p-x} = \frac{y}{y-p}.$$

Using

$$x = \frac{p(\sqrt{5}-1)}{2}, \qquad p - x = \frac{p(3-\sqrt{5})}{2},$$

we get

$$\frac{x}{p-x} = \frac{\sqrt{5}-1}{3-\sqrt{5}} = \frac{1+\sqrt{5}}{2}.$$

Hence

$$\frac{y}{y-p} = \frac{1+\sqrt{5}}{2}.$$

So

$$2y = (1+\sqrt{5})(y-p) \implies y(\sqrt{5}-1) = p(1+\sqrt{5}),$$

and therefore

$$\boxed{y = \frac{p(3+\sqrt{5})}{2}.}$$

(iii) Now

$$AB = PB - PA = y - x = \frac{p(3+\sqrt{5})}{2} - \frac{p(\sqrt{5}-1)}{2} = 2p.$$

Since $PQ = p$,

$$\boxed{AB = 2PQ.}$$

Takeaways 2.19

- The golden-ratio equation often appears as a quadratic relation in a length variable.
- Harmonic division turns a geometry problem into a ratio equation that can be solved algebraically.
- This is a good example of functions and algebra supporting classical Euclidean geometry.

Problem 2.20: The Four-Function Cycle and Composite Inverses

Let $f(x)$, $g(x)$, $h(x)$, and $t(x)$ be linear functions.

(i) Suppose that $f(x)$ and $g(x)$ are inverse functions, and $h(x)$ and $t(x)$ are also inverse functions. Let

$$P(x) = f(h(x)).$$

Prove algebraically that the inverse of $P(x)$ is

$$Q(x) = t(g(x)).$$

(ii) Find four distinct linear functions $f(x)$, $g(x)$, $h(x)$, and $t(x)$ such that

$$f(g(h(t(x)))) = x,$$

and all of the following are true:

- none of the graphs passes through the origin,
- all four graphs have different gradients,
- no two functions in your set are inverses of each other.

Hint: For part (i), use only the defining property of inverse functions and the associativity of composition. For part (ii), make the product of the four gradients equal to 1, then choose non-zero intercepts so that the overall constant term becomes 0.

Solution 2.20

(i) Since f and g are inverses, and h and t are inverses,

$$f(g(x)) = x \qquad \text{and} \qquad h(t(x)) = x.$$

Now

$$P(Q(x)) = f\big(h(t(g(x)))\big).$$

Since $h(t(g(x))) = g(x)$, we get

$$P(Q(x)) = f(g(x)) = x.$$

Hence $Q(x) = t(g(x))$ is the inverse of $P(x) = f(h(x))$.
(ii) Choose gradients

$$m_1 = 2, \qquad m_2 = 3, \qquad m_3 = \frac{1}{2}, \qquad m_4 = \frac{1}{3},$$

so that

$$m_1 m_2 m_3 m_4 = 2 \cdot 3 \cdot \frac{1}{2} \cdot \frac{1}{3} = 1.$$

Take

$$c_1 = -1, \qquad c_2 = 2, \qquad c_3 = -1, \qquad c_4 = 1.$$

Then

$$f(x) = 2x - 1, \qquad g(x) = 3x + 2, \qquad h(x) = \frac{1}{2}x - 1, \qquad t(x) = \frac{1}{3}x + 1.$$

$$h(t(x)) = \frac{1}{2}\left(\frac{1}{3}x + 1\right) - 1 = \frac{1}{6}x - \frac{1}{2},$$

$$g(h(t(x))) = 3\left(\frac{1}{6}x - \frac{1}{2}\right) + 2 = \frac{1}{2}x + \frac{1}{2},$$

$$f(g(h(t(x)))) = 2\left(\frac{1}{2}x + \frac{1}{2}\right) - 1 = x.$$

So indeed

$$\boxed{f(g(h(t(x)))) = x.}$$

Also, none passes through the origin, the gradients are all different, and

$$f^{-1}(x) = \frac{1}{2}x + \frac{1}{2} \neq h(x), \qquad g^{-1}(x) = \frac{1}{3}x - \frac{2}{3} \neq t(x).$$

Hence all conditions are satisfied.

Takeaways 2.20

- **The shoes-and-socks rule:** The inverse of a composition is the composition of the inverses in reverse order:
$$(f \circ h)^{-1} = h^{-1} \circ f^{-1}.$$
- **Associativity matters:** In general $f(g(x)) \neq g(f(x))$, but you may regroup a long composition in any convenient way.
- **University preview:** This same inverse-in-reverse-order law appears later in abstract algebra, matrix theory, and group theory.

Problem 2.21: Newton's Serpentine and the Hyperbolic Locus

Consider the family of functions

$$f(x) = \frac{2x}{x^2 + c^2},$$

where $c > 0$ is a real parameter.

(a) Determine the horizontal asymptote of $f(x)$, and find the exact point where the curve crosses this asymptote.

(b) Use calculus to find the coordinates of the local maximum and local minimum in terms of c.

(c) As c varies, find the Cartesian equation of the locus traced by the local maximum point, and state its geometric shape.

(d) By finding $f''(x)$, prove that the origin is a point of inflection for every curve in this family.

Hint: For the asymptote, compare degrees or divide top and bottom by x^2. For the turning points, use the quotient rule and set the numerator of $f'(x)$ equal to zero. In the second derivative, factor early instead of fully expanding.

Solution 2.21

(a) As $x \to \pm\infty$, $\dfrac{2x}{x^2+c^2} \to 0$, so the horizontal asymptote is

$$\boxed{y = 0}.$$

Since $f(x) = 0 \iff 2x = 0$, the curve crosses it at

$$\boxed{(0,0)}.$$

(b)

$$f'(x) = \frac{2(x^2+c^2) - (2x)(2x)}{(x^2+c^2)^2} = \frac{2(c^2-x^2)}{(x^2+c^2)^2}.$$

So $f'(x) = 0 \iff x = \pm c$. Then

$$f(c) = \frac{1}{c}, \qquad f(-c) = -\frac{1}{c}.$$

Hence the turning points are

$$\boxed{\left(c, \frac{1}{c}\right)} \text{ and } \boxed{\left(-c, -\frac{1}{c}\right)}.$$

(c) For the local maximum, $(X, Y) = \left(c, \frac{1}{c}\right)$. Eliminating c gives

$$Y = \frac{1}{X},$$

and since $c > 0$, $X > 0$. Therefore the locus is

$$\boxed{y = \frac{1}{x}\ (x > 0)},$$

the first-quadrant branch of a rectangular hyperbola.

(d) Differentiating again gives

$$f''(x) = \frac{4x(x^2-3c^2)}{(x^2+c^2)^3}.$$

So $f''(0) = 0$. Also $(x^2+c^2)^3 > 0$ for all x, and for sufficiently small $|x|$, $x^2 - 3c^2 < 0$. Hence for small $h > 0$,

$$f''(-h) > 0, \qquad f''(h) < 0,$$

so concavity changes at $x = 0$. Therefore $(0,0)$ is a point of inflection for every $c > 0$.

Takeaways 2.21

- A horizontal asymptote describes end behaviour only; a curve may still cross it at a finite point.
- The turning points $\left(c, \frac{1}{c}\right)$ and $\left(-c, -\frac{1}{c}\right)$ show a beautiful link between calculus and loci.
- Factoring early is often the cleanest way to simplify second-derivative expressions.
- This curve is a special case of Newton's serpentine family, a nice example of how classical curve theory connects with modern function study.

Problem 2.22: Inverse Rates for a Quintic Function

Consider the function

$$y = f(x) = x^5 - 1.$$

(i) Write down $\dfrac{dy}{dx}$ for the function $f(x)$.

(ii) Make x the subject (express x in terms of y) and hence find $\dfrac{dx}{dy}$.

(iii) Hence show that

$$\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$$

where both derivatives are defined.

Hint: Differentiate $y = x^5 - 1$ first. Then rearrange the equation to write x as a function of y, and differentiate this inverse rule with respect to y.

Solution 2.22

(i) Since

$$y = x^5 - 1,$$

we have

$$\frac{dy}{dx} = 5x^4.$$

(ii) Make x the subject:

$$y + 1 = x^5 \quad \Longrightarrow \quad x = (y+1)^{1/5}.$$

Therefore

$$\frac{dx}{dy} = \frac{1}{5}(y+1)^{-4/5} = \frac{1}{5(y+1)^{4/5}}.$$

(iii) Since $y + 1 = x^5$, we have

$$(y+1)^{4/5} = (x^5)^{4/5} = x^4.$$

Thus, for $x \neq 0$,

$$\frac{dx}{dy} = \frac{1}{5x^4}.$$

Hence

$$\frac{dy}{dx} \cdot \frac{dx}{dy} = 5x^4 \cdot \frac{1}{5x^4} = \boxed{1}.$$

At $x = 0$, the inverse derivative is not defined, so the product statement is only made where both derivatives exist.

Takeaways 2.22

- For inverse functions, the gradients are reciprocal where both gradients exist and the original gradient is non-zero.
- The identity $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ is powerful, but it must be used with attention to points where a derivative is zero or undefined.
- This is mainly an Extension 2 calculus idea, but it connects naturally with inverse functions from Extension 1.

Remark 2.1

Let $y = f(x)$ be a real-valued function that is continuous and strictly monotonic on an interval containing x_0 (which guarantees the existence of an inverse function, $x = f^{-1}(y)$). If $f(x)$ is differentiable at x_0 and $f'(x_0) \neq 0$, then the inverse function is differentiable at $y_0 = f(x_0)$, and:

$$\left.\frac{dy}{dx}\right|_{x=x_0} \cdot \left.\frac{dx}{dy}\right|_{y=y_0} = 1$$

This theorem can be rigorously proved or demonstrated using:

1. The Limit Definition of the Derivative (First Principles)
2. The Geometric Interpretation of Gradients
3. The Chain Rule

3 Part 2: Practice and Challenge Bank

3.1 Warm-Up Drills

Problem 3.1: Evaluation & Domains

Find the maximal natural domain and range of the following functions:

(i) $f(x) = \sqrt{4 - x}$

(ii) $g(x) = \frac{1}{x+2}$

Hint: For square roots, the inside must be non-negative. For fractions, the denominator cannot be zero.

Solution 3.1

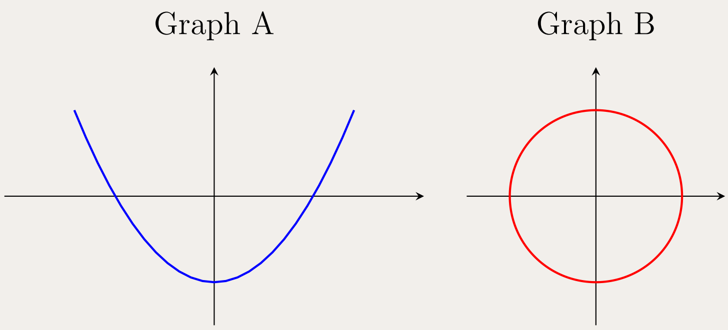
(i) We require $4 - x \geq 0 \implies x \leq 4$. Domain: $(-\infty, 4]$. Since the principal square root is non-negative, Range: $[0, \infty)$.
(ii) We require $x + 2 \neq 0 \implies x \neq -2$. Domain: $\mathbb{R} \setminus \{-2\}$. Since the numerator is 1, $g(x)$ can never be 0. Range: $\mathbb{R} \setminus \{0\}$.

Takeaways 3.1

- Check for division by zero and square roots of negative numbers to find the domain.
- Realise what outputs are impossible (like 0 in a basic rational function) to establish the range.

Problem 3.2: Function Tests

Below are two graphs. Use the vertical and horizontal line tests to determine which graph represents a valid function, and whether it is a one-to-one function.



Hint: If any vertical line crosses the graph more than once, it's not a function. If any horizontal line crosses it more than once, it is not one-to-one.

Solution 3.2

Graph A: A parabola opening upwards. It passes the vertical line test, so it is a valid function. However, any horizontal line above the vertex intersects it twice, so it is **not** one-to-one.
Graph B: A circle. A vertical line (e.g., the y-axis) intersects the curve at two points. Therefore, Graph B does **not** represent a valid function.

Takeaways 3.2

- Vertical line test confirms $y = f(x)$ is single-valued mapping.
- Horizontal line test confirms injectivity (one-to-one).

Problem 3.3: Parity of Functions

Determine algebraically whether the following functions are even, odd, or neither:

(i) $f(x) = x^3 - x$

(ii) $g(x) = x^2 \cos(x)$

Hint: Substitute $-x$ into the function. If it yields exactly $f(x)$, it's even. If it yields $-f(x)$, it's odd.

Solution 3.3

(i) Substitute $-x$: $f(-x) = (-x)^3 - (-x) = -x^3 + x = -(x^3 - x) = -f(x)$. Because $f(-x) = -f(x)$, the function $f(x)$ is **odd**.
(ii) Substitute $-x$: $g(-x) = (-x)^2 \cos(-x) = x^2 \cos(x) = g(x)$ (since cosine is an even function). Because $g(-x) = g(x)$, the function $g(x)$ is **even**.

Takeaways 3.3

- Always simplify carefully, paying attention to the signs on odd and even powers.
- Standard trig functions have known parities: $\cos x$ is even, $\sin x$ is odd.

Problem 3.4: Basic Composition

Given the functions $f(x) = 2x + 1$ and $g(x) = x^2$, find the rules for:

(i) $f(g(x))$

(ii) $g(f(x))$

Hint: Substitute the entire expression of one function into the input variable of the other.

Solution 3.4

(i) Replace x in f with $g(x)$:

$$f(g(x)) = 2(x^2) + 1 = \boxed{2x^2 + 1}.$$

(ii) Replace x in g with $f(x)$:

$$g(f(x)) = (2x + 1)^2 = \boxed{4x^2 + 4x + 1}.$$

Takeaways 3.4

- Function composition is generally **not** commutative: $f(g(x)) \neq g(f(x))$.
- Keep parentheses around substituted expressions to prevent expansion errors.

Problem 3.5: Parity of Composite Sine Functions

Consider the function $f(x) = \sin(x^5 + x)$. Which of the following statements regarding the parity of $f(x)$ is correct?
A) $f(x)$ is an even function because the graph of a sine wave is symmetric.
B) $f(x)$ is an even function because $(-x)^5 + (-x)$ becomes positive when squared.
C) $f(x)$ is an odd function because both the inner function and the sine function are odd.
D) $f(x)$ is neither even nor odd.

Hint: Inner Parity: Check $u(x) = x^5 + x$ again. Is $u(-x)$ equal to $u(x)$ or $-u(x)$? Recall the identity for sine. If you have an odd function inside another odd function, what happens?

Solution 3.5

The correct answer is C.
1. Test the inner function:
Let $u(x) = x^5 + x$.
$u(-x) = (-x)^5 + (-x) = -(x^5 + x) = -u(x)$.
The inner function is **odd**.
2. Apply the outer function:
$f(-x) = \sin(u(-x)) = \sin(-u(x))$.
Using the property that sine is an odd function:
$\sin(-u(x)) = -\sin(u(x)) = -f(x)$.
3. Conclusion:
Since $f(-x) = -f(x)$, the function is **odd**.

Takeaways 3.5

- **Odd + Odd = Odd:** When an odd function is composed with another odd function, the result remains odd.
- **Contrast with Cosine:** This is why $\cos(x^5 + x)$ was even but $\sin(x^5 + x)$ is odd. Cosine "swallows" the negative sign; Sine "spits it out."
- **Visual Check:** Odd functions have **rotational symmetry** of 180° about the origin.

Problem 3.6: The Structural Property of Functional Associativity

Prove that function composition is always associative. That is, for any three functions f, g, and h where the compositions are well-defined, show that:

$$((f \circ g) \circ h)(x) = (f \circ (g \circ h))(x)$$

for all x in the common domain.

Hint: Treat $(f \circ g)$ as a single composite function, say $u(x)$, and expand the LHS. Do the same for $(g \circ h)$ on the RHS. Your objective is to show that both sides collapse into $f(g(h(x)))$.

Solution 3.6

1. Analyze the Left-Hand Side (LHS):
By the definition of composition, we treat the grouped term $(f \circ g)$ as the outer function:

$$((f \circ g) \circ h)(x) = (f \circ g)(h(x))$$

Applying the definition again to the term $(f \circ g)$ with the input $h(x)$:

$$(f \circ g)(h(x)) = f(g(h(x)))$$

2. Analyze the Right-Hand Side (RHS):
By the definition of composition, we treat the grouped term $(g \circ h)$ as the inner function:

$$(f \circ (g \circ h))(x) = f((g \circ h)(x))$$

Applying the definition again to the term $(g \circ h)(x)$ inside the brackets:

$$f((g \circ h)(x)) = f(g(h(x)))$$

Conclusion:
Since both expressions evaluate to the same sequential mapping $f(g(h(x)))$, the identity holds:

$$((f \circ g) \circ h)(x) = (f \circ (g \circ h))(x)$$

Takeaways 3.6

- **The Sequential Nature of Functions:** This proof demonstrates that function composition is fundamentally about the **order of application**.
- **Mapping Chains:** You can visualize this as a chain of sets. The associativity simply confirms that the "path" through these sets remains unchanged regardless of how we algebraically "chunk" the steps.
- **Abstract Rigor:** We did not need to know what f, g, or h actually *do*. The proof relies solely on the **structure** of the operation.

Problem 3.7: Functional Dilation and Stretching

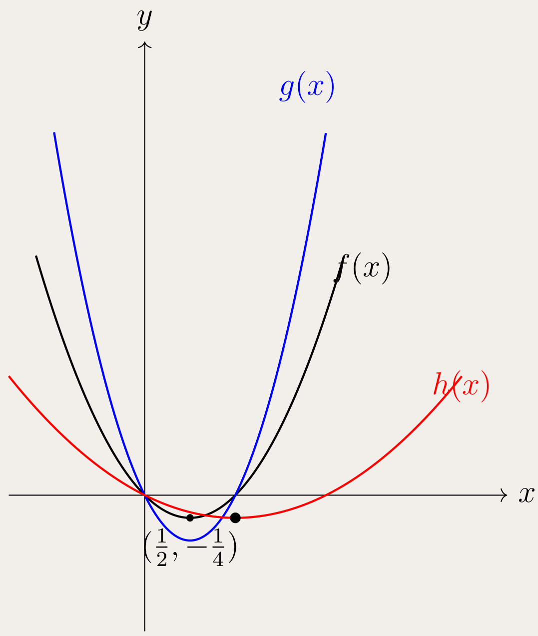
Consider the base function $f(x) = x(x-1)$. The following two functions are defined as transformations of $f(x)$:

$$g(x) = 2f(x) \quad \text{and} \quad h(x) = f\left(\frac{x}{2}\right)$$

(i) Sketch $y = f(x)$ on a Cartesian plane, clearly labeling the x-intercepts and the stationary point (vertex).

(ii) On the same diagram, sketch $y = g(x)$. Geometrically describe the dilation from $f(x)$ to $g(x)$. State any points that remain invariant under this transformation.

(iii) On the same diagram, sketch $y = h(x)$. Define $h(x)$ algebraically and calculate the coordinates of its stationary point.



Hint: For $g(x)$, every y-value is doubled while the x-values remain constant. For $h(x)$, the input variable is "halved", so to achieve the same output, x must be doubled (horizontal stretch).

Solution 3.7

(i) Base Function:
$f(x) = x^2 - x$. Intercepts at $(0, 0)$ and $(1, 0)$. Vertex at $(0.5, -0.25)$.
(ii) Vertical Dilation:
$g(x) = 2x^2 - 2x$. This is a **vertical dilation** from the x-axis by a factor of 2. Invariant points are the intercepts $(0, 0)$ and $(1, 0)$ because their y-values are zero.
(iii) Horizontal Dilation:
$h(x) = \frac{x^2}{4} - \frac{x}{2}$. This is a **horizontal dilation** from the y-axis by a factor of 2. The y-value of the vertex remains -0.25, but the x-coordinate is dilated from 0.5 to 1. New vertex at $(1, -0.25)$.

Takeaways 3.7

- **Direction of Change:** Changes outside the function (like $2f(x)$) affect the y-axis (vertical). Changes inside the function (like $f(x/2)$) affect the x-axis (horizontal).
- **The Scale Factor Paradox:** Note that in $f(x/k)$, the horizontal stretch factor is actually k. Students often mistakenly think $f(x/2)$ is a compression, but it is actually a stretch.

3.2 Stretch Problems

Problem 3.8: Domain Restriction for Inverse

Consider the quadratic function $f(x) = x^2 - 4x + 3$.

(i) Determine the vertex and sketch the graph.

(ii) Restrict the domain of $f(x)$ minimally such that it becomes a one-to-one function including $x = 5$.

(iii) Find the rule and domain for the inverse function $f^{-1}(x)$ for your restriction.

Hint: Complete the square to find the vertex. For the inverse to exist, you need to pick a branch (left or right of the vertex) that includes $x = 5$. Then swap x and y and solve for y.

Solution 3.8

(i) Completing the square gives $f(x) = (x-2)^2 - 1$. The vertex is $(2, -1)$.
(ii) To make $f(x)$ one-to-one and include $x = 5$, we must restrict the domain to the right branch originating from the vertex. The restricted domain is $[2, \infty)$.
(iii) Let $y = (x-2)^2 - 1$, where $x \geq 2$, so $y \geq -1$. Swap x and y to find the inverse:

$$x = (y-2)^2 - 1 \implies x + 1 = (y-2)^2.$$

Since the original domain had $x \geq 2$, our new y must be ≥ 2, so we take the positive square root:

$$y - 2 = \sqrt{x+1} \implies y = 2 + \sqrt{x+1}.$$

Rule: $\boxed{f^{-1}(x) = 2 + \sqrt{x+1}}$. Domain of f^{-1} is the range of restricted f: $\boxed{[-1, \infty)}$.

Takeaways 3.8

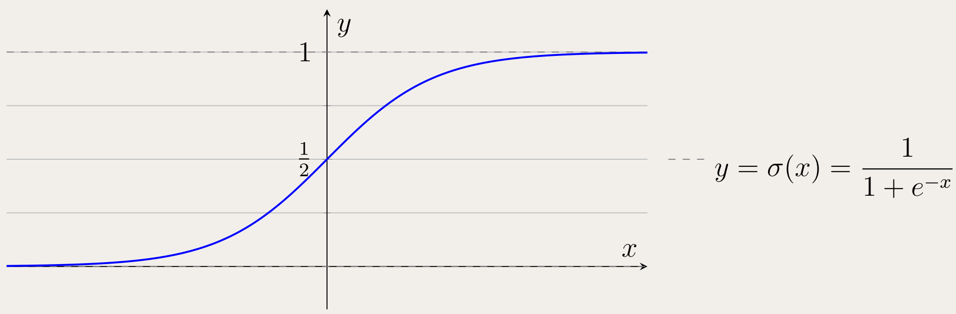
- Quadratics must be split at the vertex to secure a one-to-one mapping.
- Selecting the $\pm$ sign when taking the square root determines which branch of the parabola the inverse describes.

Problem 3.9: The Sigmoid Function and Its Inverse

In artificial intelligence, the Sigmoid function is widely used as an activation function in neural networks to map any real-valued input into a strictly bounded output between 0 and 1. The standard Sigmoid function is defined by

$$\sigma(x) = \frac{1}{1+e^{-x}}.$$

The graph of $\sigma(x)$ is an S-shaped curve that approaches 0 as $x \to -\infty$ and approaches 1 as $x \to \infty$, as shown below.



(i) Using the rules of differentiation, prove that

$$\sigma'(x) = \sigma(x)\big(1 - \sigma(x)\big).$$

(ii) Hence, or otherwise, determine the exact coordinates of the point of inflection (i.e. the point where the curve changes concavity, or in other words, where the second derivative equals zero) of the curve $y = \sigma(x)$.

(iii) Find an algebraic expression for the inverse function $\sigma^{-1}(x)$ and state its domain.

Hint:

- For (i), rewrite $\sigma(x) = (1+e^{-x})^{-1}$ and apply the chain rule. Then compute $1-\sigma(x)$ and compare.
- For (ii), the inflection point occurs where $\sigma''(x) = 0$. Differentiate $\sigma'(x) = \sigma(x)\big(1-\sigma(x)\big)$ using the product rule.
- For (iii), set $y = \sigma(x)$, swap x and y, and solve for y using logarithms. The domain of σ^{-1} is the range of σ.

Solution 3.9

(i) Write $\sigma(x) = (1+e^{-x})^{-1}$. Then by the chain rule,

$$\sigma'(x) = -(1+e^{-x})^{-2} \cdot (-e^{-x}) = \frac{e^{-x}}{(1+e^{-x})^2}.$$

Also,

$$1 - \sigma(x) = 1 - \frac{1}{1+e^{-x}} = \frac{e^{-x}}{1+e^{-x}},$$

so

$$\sigma(x)\big(1-\sigma(x)\big) = \frac{1}{1+e^{-x}} \cdot \frac{e^{-x}}{1+e^{-x}} = \frac{e^{-x}}{(1+e^{-x})^2} = \sigma'(x).$$

(ii) Let $y = \sigma(x)$. From (i), $y' = y(1-y)$. Differentiate:

$$y'' = y'(1-y) - yy' = y'(1-2y).$$

Since $y' = \dfrac{e^{-x}}{(1+e^{-x})^2} > 0$ for all x, the inflection point satisfies $1-2y = 0$, so $y = \frac{1}{2}$. Then

$$\frac{1}{2} = \frac{1}{1+e^{-x}} \implies 1 + e^{-x} = 2 \implies e^{-x} = 1 \implies x = 0.$$

So the point of inflection is $\boxed{\left(0, \dfrac{1}{2}\right)}$.

(iii) Let $y = \dfrac{1}{1+e^{-x}}$. Then

$$\frac{1}{y} = 1 + e^{-x} \implies e^{-x} = \frac{1-y}{y}.$$

Taking ln,

$$-x = \ln\left(\frac{1-y}{y}\right) \implies x = \ln\left(\frac{y}{1-y}\right).$$

Hence

$$\boxed{\sigma^{-1}(x) = \ln\left(\frac{x}{1-x}\right)}$$

with domain $\boxed{x \in (0,1)}$.

Takeaways 3.9

- **Self-referential derivative:** The identity $\sigma' = \sigma(1-\sigma)$ is a classic example where the derivative can be written purely in terms of the original function.
- **Inflection from structure:** Once $\sigma' = \sigma(1-\sigma)$ is known, σ'' factors cleanly, making the inflection point quick to locate.
- **Logit link:** The inverse $\sigma^{-1}(x) = \ln\left(\dfrac{x}{1-x}\right)$ (the *logit* function) naturally has domain $(0,1)$, matching probability outputs.

Problem 3.10: Piecewise Functions

A function is defined as:

$$h(x) = \begin{cases} 2x+3, & \text{if } x < -1 \\ x^2, & \text{if } x \geq -1 \end{cases}$$

Find the range of $h(x)$ and determine whether the function is continuous at the break point $x = -1$.

Hint: Sketch the function or analyze the output of each piece. For continuity, check if the limits from the left and right match the function's value at $x = -1$.

Solution 3.10

Range: For $x < -1$, the graph is a line $y = 2x + 3$ with an open endpoint at $(-1, 1)$. Thus, this piece sweeps out values in $(-\infty, 1)$. For $x \geq -1$, the graph is a parabola $y = x^2$ restricted to $x \geq -1$. The minimum is at $(0, 0)$, and at $x = -1$, $y = 1$. This piece sweeps out values in $[0, \infty)$. The union of $(-\infty, 1)$ and $[0, \infty)$ is all real numbers. Thus, the range is $\boxed{\mathbb{R}}$.

Continuity: Left limit as $x \to -1^-$ uses the rule $2x + 3 \to 1$. Right limit as $x \to -1^+$ uses the rule $x^2 \to 1$. The function value at $x = -1$ is $(-1)^2 = 1$. Since all three agree, the function is $\boxed{\text{continuous}}$ at $x = -1$.

Takeaways 3.10

- The overall range of a piecewise function is the union of the ranges of its constituent parts.
- Ensuring left and right limits equate at break points is the core test for continuity.

Problem 3.11: Abstract Transformations

Given the graph of a generic function $y = f(x)$, describe the sequence of geometric transformations required to obtain the graph of:

(i) $y = 2f(1 - x) - 3$

(ii) Describe the difference in graphing $y = |f(x)|$ compared to $y = f(|x|)$.

Hint: For part (i), rewrite the input as $-(x - 1)$. For part (ii), think about which parts of the inputs/outputs are forced to be positive.

Solution 3.11

(i) Rewrite the equation as $y = 2f(-(x - 1)) - 3$. The transformations from $y = f(x)$ are: 1. Reflection across the y-axis: $f(-x)$ 2. Horizontal translation right by 1 unit: $f(-(x - 1))$ 3. Vertical dilation by a factor of 2: $2f(1 - x)$ 4. Vertical translation down by 3 units: $2f(1 - x) - 3$

(ii) To graph $y = |f(x)|$, apply the absolute value to the outputs: take any part of the graph below the x-axis and reflect it up across the x-axis. To graph $y = f(|x|)$, apply the absolute value to the inputs: discard the left side of the graph ($x < 0$), keep the right side ($x \geq 0$), and reflect the right side across the y-axis to create an even function.

Takeaways 3.11

- Factoring out coefficients in the input parameter prevents translation errors (e.g., $f(1-x) \to f(-(x-1))$).
- Absolute value outside affects y-values (folding up), inside affects x-values (mirroring right to left).

Problem 3.12: Rational Decomposition and Asymptotic Behavior

Consider the rational function $f(x) = \frac{x^2-x+2}{x-2}$.

(i) Use polynomial division to show that:

$$f(x) = x + 1 + \frac{4}{x-2}$$

(ii) Deduce that $y = x + 1$ is an oblique asymptote for the graph of $y = f(x)$. Explain the behavior of the function as $x \to \infty$ in terms of this line.

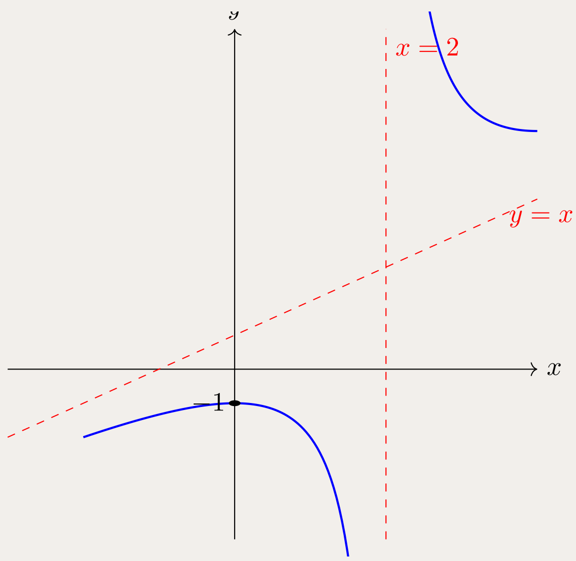
(iii) State the equation of the vertical asymptote and find the coordinates of the y-intercept.

(iv) Sketch the graph of $y = f(x)$, showing all asymptotes and intercepts.

Hint: When dividing, the quotient represents the linear part of the asymptote, and the remainder forms the fractional part that vanishes at infinity. Consider the limit as $x \to \pm\infty$.

Solution 3.12

(i) Division:
Dividing $x^2 - x + 2$ by $x - 2$:
Quotient is $x + 1$, remainder is 4. Thus, $f(x) = x + 1 + \frac{4}{x-2}$.
(ii) Asymptote Logic:
As $x \to \pm\infty$, the term $\frac{4}{x-2} \to 0$.
Therefore, $f(x) \approx x + 1$ for very large positive or negative values of x. The line $y = x + 1$ is the oblique asymptote.
(iii) Intercepts and Asymptotes:
Vertical Asymptote: $x = 2$.
y-intercept: $f(0) = \frac{0-0+2}{0-2} = -1$, yielding point $(0, -1)$.
(iv) Sketching:
The graph is a hyperbola-like shape. To the right of $x = 2$, the curve is above the asymptote ($x + 1 +$ positive). To the left, it is below.



Takeaways 3.12

- **Decomposition is Key:** Writing a rational function as Polynomial+Proper Fraction immediately reveals its "end-game" behavior.
- **Oblique vs. Horizontal:** An oblique asymptote occurs when the degree of the numerator is exactly **one higher** than the degree of the denominator.

Problem 3.13: The Geometry of the Modulus Plateau

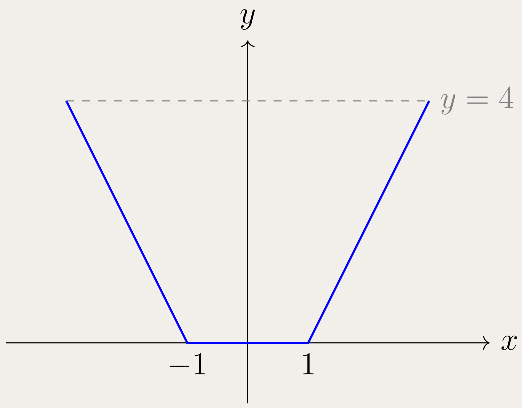
Analyze the function $f(x) = |x+1| + |x-1| - 2$.

(i) Define $f(x)$ as a piecewise function for the intervals $x < -1$, $-1 \leq x \leq 1$, and $x > 1$.

(ii) Sketch the graph of $y = f(x)$. Describe the behavior of the function on the interval $[-1, 1]$.

(iii) Solve the inequality $|x+1| + |x-1| - 2 \leq 4$ using your graph.

(iv) Calculate the definite integral $\int_{-2}^{2} f(x)\, dx$.



Hint: The region between the critical points often results in the variable x canceling out, creating a horizontal line segment. For the integral, use the geometry of the area under the curve.

Solution 3.13

(i) Piecewise Definition:
$x < -1$**:** $f(x) = -(x+1) - (x-1) - 2 = -2x - 2$.
$-1 \leq x \leq 1$**:** $f(x) = (x+1) - (x-1) - 2 = 0$.
$x > 1$**:** $f(x) = (x+1) + (x-1) - 2 = 2x - 2$.
(ii) The Graph:
The graph is "V-shaped" but with a flat base. It sits flat on the x-axis between $x = -1$ and $x = 1$.
(iii) Inequality:
To solve $f(x) \leq 4$: On the right branch: $2x - 2 = 4 \implies x = 3$. On the left branch: $-2x - 2 = 4 \implies x = -3$. The solution is $-3 \leq x \leq 3$.
(iv) Definite Integral:
From $x = -1$ to $x = 1$, the area is 0. From $x = 1$ to $x = 2$, $f(2) = 2$. The area is a triangle: $\frac{1}{2} \times 1 \times 2 = 1$. Total integral is 2 by symmetry.

Takeaways 3.13

- **The Plateau Effect:** When you sum two modulus terms with the same coefficient, the "inner" region always results in a constant value.
- **Critical Point Analysis:** Most complex modulus problems are solved by simply "breaking" the number line at the points where the behavior changes.

Problem 3.14: Inequalities of Sine Reciprocals

Consider the equation: $\dfrac{1}{1+\sin x} = 1 - \dfrac{x}{\pi}$

(i) Verify that $x = 0$ and $x = \frac{\pi}{2}$ satisfy the equation.

(ii) Sketch $y = 1 - \frac{x}{\pi}$ and $y = \frac{1}{1+\sin x}$ for $0 \le x \le \pi$.

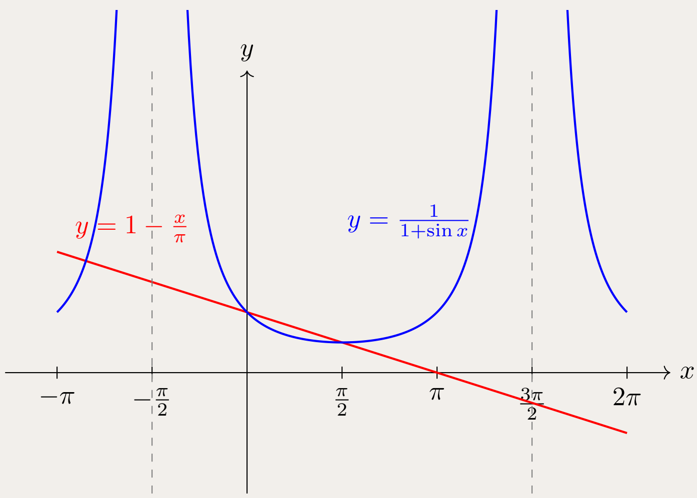
(iii) Using the relative positions of the graphs, show that $\dfrac{1}{1+\sin x} < \dfrac{\pi - x}{\pi}$ for $0 < x < \frac{\pi}{2}$.

(iv) Hence, prove that $\left(\sin\frac{x}{2} + \cos\frac{x}{2}\right)^2 > \dfrac{\pi}{\pi - x}$ for $0 < x < \frac{\pi}{2}$.

Hint: The curve $y = \frac{1}{1+\sin x}$ starts at $(0, 1)$, dips to $(\frac{\pi}{2}, \frac{1}{2})$, and rises back to $(\pi, 1)$. For part (iv), take the reciprocal of both sides (flipping the inequality sign).

Solution 3.14

(i) Verification:
At $x = 0$: LHS $= \frac{1}{1+0} = 1$; RHS $= 1 - 0 = 1$. Correct.
At $x = \frac{\pi}{2}$: LHS $= \frac{1}{1+1} = \frac{1}{2}$; RHS $= 1 - \frac{\pi/2}{\pi} = 1 - \frac{1}{2} = \frac{1}{2}$. Correct.
(ii) The Sketch:
The curve is a "valley" shape. The line is a simple downward slope. Between $x = 0$ and $x = \pi/2$, the curve drops faster than the line.



(iii) Inequality:
Since the curve lies below the line on that interval:
$$\frac{1}{1+\sin x} < \frac{\pi - x}{\pi}$$
(iv) Derivation:
Take the reciprocal of both sides: $1 + \sin x > \frac{\pi}{\pi - x}$.
Use the identity $1 + \sin x = (\sin\frac{x}{2} + \cos\frac{x}{2})^2$. Substituting this yields the desired result.

Takeaways 3.14

- **Linear Approximation:** Problems like this show how trig functions can be bounded by straight lines.
- **The Power of $1 + \sin x$:** This denominator is a favorite in Extension 2 because it can be rewritten as a perfect square: $(\sin \frac{x}{2} + \cos \frac{x}{2})^2$.

Problem 3.15: The Rotated Modulus Region

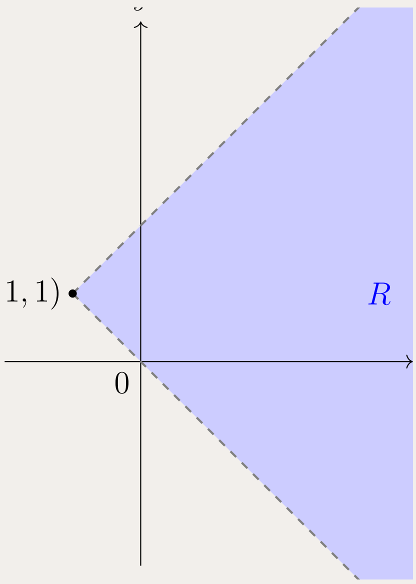
Sketch and define the region R in the Cartesian plane specified by the inequality $x + 1 > |y - 1|$.

(i) Describe the boundary of the region and find the coordinates of its "vertex."

(ii) Determine whether the origin $(0, 0)$ is an interior point of the region R.

(iii) Write the definition of the region R using a system of two piecewise linear inequalities.

(iv) Sketch the region R, clearly showing asymptotes, intercepts, and vertex.



Hint: The shape is not a parabola but a "V" shape rotated 90° clockwise. Test $(0, 0)$ to see if it makes the inequality true.

Solution 3.15

(i) The Boundary:
The boundary is $x = |y-1| - 1$. This is a V-shaped graph shifted left 1 unit and up 1 unit. Vertex: $(-1, 1)$.
(ii) Test Point:
Substitute $(0,0)$: $0+1 > |0-1| \implies 1 > 1$. This is false. Therefore, the origin $(0,0)$ is not in the region R.
(iii) Piecewise Inequalities:
For $y \geq 1$: $x > (y-1) - 1 \implies x > y - 2$.
For $y < 1$: $x > -(y-1) - 1 \implies x > -y$.
Combined: $R = \{x > y - 2\} \cap \{x > -y\} \cap \{y \neq 1\}$.

Takeaways 3.15

- **Rotated Perspective:** The absolute value on the y-variable means the "sharp point" is oriented horizontally.
- **Region Logic:** Visually, $y > |x|$ is "above the V," whereas $x > |y|$ is "to the right of the V."

Problem 3.16: Composite and Inverse Domains

The functions f and g are defined by $f(x) = \sqrt{3x-2}$ and $g(x) = x + 1$ over their maximal real domains.

(i) Find the expression for the composite function $h(x) = (f \circ g)(x)$ and state its maximal domain.

(ii) Find the inverse function $h^{-1}(x)$ and state its domain.

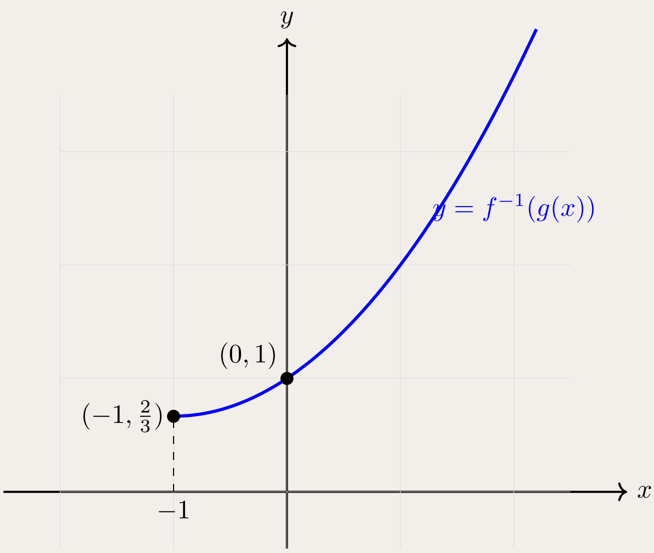
(iii) Sketch the graph of $y = f^{-1}(g(x))$, clearly labeling the coordinates of the endpoint and the y-intercept.

Hint:

- **Part (i):** Substitute $g(x)$ into $f(x)$. To find the domain of the composite function, consider the restriction caused by the square root.
- **Part (ii):** Let $y = h(x)$ and swap x and y. Remember that the domain of an inverse function $h^{-1}(x)$ is exactly the range of the original function $h(x)$.
- **Part (iii):** First, find the explicit equation for $f^{-1}(x)$ and note its domain. When substituting $g(x)$ into $f^{-1}(x)$, the outputs of $g(x)$ must fall within the domain of f^{-1}. This will restrict the x-values you are allowed to plot.

Solution 3.16

(i)
$h(x) = f(g(x))$
$h(x) = \sqrt{3(x+1)-2}$
$h(x) = \sqrt{3x+1}$
For the maximal domain, the expression under the square root must be non-negative:
$3x+1 \geq 0 \implies x \geq -\frac{1}{3}$
Domain: $x \geq -\frac{1}{3}$
(ii)
Let $y = \sqrt{3x+1}$. The range of $h(x)$ is $y \geq 0$.
To find the inverse, swap x and y:
$x = \sqrt{3y+1}, \quad \text{where } x \geq 0$
$x^2 = 3y+1$
$3y = x^2 - 1$
$y = \frac{x^2-1}{3}$
Therefore, $h^{-1}(x) = \frac{x^2-1}{3}$.
Domain: $x \geq 0$ (since the domain of h^{-1} is the range of h).
(iii)
First, find $f^{-1}(x)$.
$f(x) = \sqrt{3x-2}$. Its domain is $x \geq \frac{2}{3}$ and its range is $y \geq 0$.
Swap x and y: $x = \sqrt{3y-2}$ for $x \geq 0$.
$x^2 = 3y - 2 \implies y = \frac{x^2+2}{3}$.
So, $f^{-1}(x) = \frac{x^2+2}{3}$ with domain $x \geq 0$.
Now, find $y = f^{-1}(g(x))$:
$y = f^{-1}(x+1) = \frac{(x+1)^2+2}{3}$
Domain Restriction: The outer function f^{-1} only accepts inputs ≥ 0. Therefore, $g(x) \geq 0$.
$x+1 \geq 0 \implies x \geq -1$.
The graph is the right half of a parabola starting from the vertex at $x=-1$.
Endpoint: $(-1, \frac{2}{3})$. y-intercept: $(0,1)$.



Takeaways 3.16

- **Composite Domains:** The domain of a composite function $f(g(x))$ requires the inner function $g(x)$ to be defined, AND the output of $g(x)$ to fall within the domain of the outer function $f(x)$.
- **Inverse Domains:** Always state the domain and range of the original function *before* finding the inverse. Mapping Domain$(f) \to$ Range(f^{-1}) is the most robust way to avoid losing marks.
- **Hidden Restrictions:** In part (iii), algebraic substitution yields a full parabola, but the functional definition restricts it to half a parabola. Always verify the domain of the outer function when composing.

3.3 Challenge Corner

Problem 3.17: Self-Inverse Rational Functions

Prove that any rational function of the form

$$f(x) = \frac{ax+b}{cx-a}$$

where a, b, c are constants and $a^2 + bc \neq 0$, is its own inverse.

Hint: Set $y = f(x)$, swap x and y, and algebraically solve for the new y. What happens when $a^2 + bc = 0$?

Solution 3.17

Let $y = \frac{ax+b}{cx-a}$. Swap variables to find $f^{-1}(x)$:

$$x = \frac{ay+b}{cy-a}.$$

Solve for y:

$$\begin{aligned} x(cy-a) &= ay+b \\ cxy - ax &= ay + b \\ cxy - ay &= ax + b \\ y(cx-a) &= ax+b \\ y &= \frac{ax+b}{cx-a}. \end{aligned}$$

Since y evaluates back to the exact same expression as $f(x)$, we have $f^{-1}(x) = f(x)$. Hence, the function is its own inverse. (Note: $a^2 + bc \neq 0$ is required so the function is not just a constant, ensuring it is a one-to-one function.)

Takeaways 3.17

- A function that equals its own inverse is called an involution.
- Geometrically, $f(f(x)) = x$ implies the graph is perfectly symmetrical across the line $y = x$.

Problem 3.18: Linear Function Composition

Find all linear functions $f(x) = mx + c$ such that $f(f(x)) = 4x - 9$.

Hint: Compute $f(f(x))$ algebraically without replacing constants, then expand and equate coefficients with $4x - 9$.

Solution 3.18

Given $f(x) = mx + c$, evaluate $f(f(x))$:

$$f(f(x)) = m(mx + c) + c = m^2x + mc + c.$$

We are given that $f(f(x)) = 4x - 9$. Equating coefficients: 1) $m^2 = 4 \implies m = 2$ or $m = -2$. 2) $mc + c = -9 \implies c(m + 1) = -9$.

Case 1: $m = 2$. $c(2 + 1) = -9 \implies 3c = -9 \implies c = -3$. This yields the function $\boxed{f(x) = 2x - 3}$.

Case 2: $m = -2$. $c(-2 + 1) = -9 \implies -c = -9 \implies c = 9$. This yields the function $\boxed{f(x) = -2x + 9}$.

Takeaways 3.18

- Equating coefficients is a powerful technique whenever polynomial degrees match.
- Remembering that quadratic equations ($m^2 = 4$) yield dual solutions protects against losing potential answers.

Problem 3.19: Advanced Composition Domain

Find the valid geometric domain of the composite function $f(g(x))$ where $f(x) = \sqrt{x}$ and $g(x) = 1 - x^2$.

Hint: The domain of a composite function comprises the restrictions of the inner function, further constrained so the output of the inner function fits within the domain of the outer function.

Solution 3.19

For $f(g(x))$ to exist: 1. x must be in the domain of $g(x)$. The domain of $g(x) = 1 - x^2$ is $\mathbb{R}$. 2. The output $g(x)$ must lie within the domain of $f(x)$, which is $x \geq 0$. So we require $g(x) \geq 0$:

$$1 - x^2 \geq 0 \implies x^2 \leq 1 \implies -1 \leq x \leq 1.$$

Comparing the two conditions, the overall domain is the intersection, which is simply $\boxed{[-1, 1]}$.

Takeaways 3.19

- It often helps to evaluate $\text{dom}(f \circ g) = \{x \in \text{dom}(g) \mid g(x) \in \text{dom}(f)\}$.
- Even if the simplified formula $\sqrt{1 - x^2}$ implies a domain, relying on checking the intersection of inner outputs with outer domains avoids errors with more convoluted functions.

Problem 3.20: Investigation of $f(x) = |x+1| + \frac{1}{x-1}$

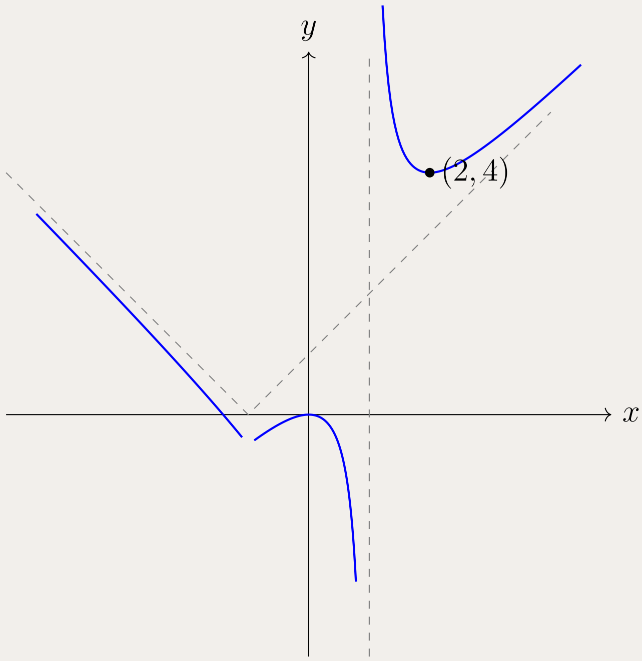
Analyze the function $f(x) = |x+1| + \frac{1}{x-1}$ for $x \neq 1$.

(i) Define $f(x)$ as a piecewise function for the intervals $(-\infty, -1)$, $[-1, 1)$, and $(1, \infty)$.

(ii) State the equations of all asymptotes.

(iii) Find the coordinates of the local minimum for $x > 1$.

(iv) Sketch the graph of the function.



Hint: The modulus $|x+1|$ flips at $x = -1$, but the vertical asymptote lies at $x = 1$. Differentiate for $x > 1$ ignoring the absolute value sign since $x + 1 > 0$.

Solution 3.20

(i) Piecewise Definition:
$x < -1$: $f(x) = -(x+1) + \frac{1}{x-1}$
$-1 \leq x < 1$: $f(x) = (x+1) + \frac{1}{x-1}$
$x > 1$: $f(x) = (x+1) + \frac{1}{x-1}$
(ii) Asymptotes:
Vertical: $x = 1$.
Oblique ($x \to \infty$): $y = x + 1$.
Oblique ($x \to -\infty$): $y = -x - 1$.
(iii) Stationary Point ($x > 1$):
$f'(x) = 1 - \frac{1}{(x-1)^2}$.
Set $f'(x) = 0 \implies (x-1)^2 = 1 \implies x - 1 = 1 \implies x = 2$. $f(2) = |2+1| + \frac{1}{2-1} = 4$. Local minimum at $(2, 4)$.

Takeaways 3.20

- **Independent Shifts:** The V-shape vertex was shifted left to $x = -1$, while the vertical asymptote was shifted right to $x = 1$.
- **Asymptote Interaction:** The oblique asymptotes are no longer symmetric about the y-axis; they are symmetric about the line $x = -1$.

Problem 3.21: The Hyperbolic Divide

Investigate the region R defined by the simultaneous inequalities:

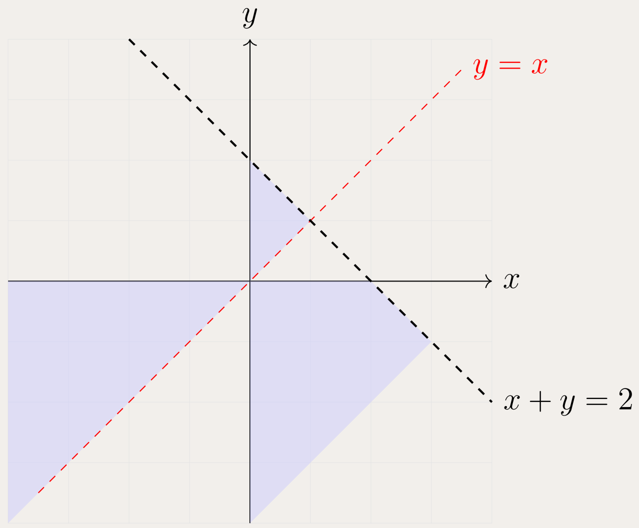
$$\frac{1}{x} > \frac{1}{y} \quad \text{and} \quad x + y < 2$$

(i) Show that the inequality $\frac{1}{x} > \frac{1}{y}$ is equivalent to $\frac{y-x}{xy} > 0$.

(ii) Identify the regions in the four quadrants that satisfy $\frac{y-x}{xy} > 0$.

(iii) Sketch the region R, showing the boundary lines $y = x$, $x + y = 2$, and the coordinate axes.

(iv) Algebraically determine if the point $(1, 0.5)$ lies within the region R.



Hint: You cannot simply cross-multiply. Instead, move both terms to one side. The "correct" region will be the intersection of the hyperbolic zones and the area "below" the line.

Solution 3.21

(i) Algebraic Logic:
$\frac{1}{x} - \frac{1}{y} > 0 \implies \frac{y-x}{xy} > 0.$
(ii) Quadrant Analysis:
Q1 $(x, y > 0)$: Satisfied when $y > x$.
Q2 $(x < 0, y > 0)$: Impossible, as y is pos and x is neg.
Q3 $(x, y < 0)$: Satisfied when $y > x$ (e.g., $-1 > -2$).
Q4 $(x > 0, y < 0)$: Always true.
(iii) Intersection:
The line $x + y = 2$ intersects $y = x$ at $(1, 1)$. The region R consists of the triangle in Q1 bounded by $y = x$, $x + y = 2$, and the y-axis; all of Q4 below the line $x + y = 2$; all of Q3 where $y > x$.
(iv) Test Point:
At $(1, 0.5)$: $\frac{1}{1} > \frac{1}{0.5} \implies 1 > 2$ (FALSE). The point does not lie in the region.

Takeaways 3.21

- **The Danger of Cross-Multiplication:** In inequalities, multiplying by a variable is dangerous because you don't know if you need to flip the inequality sign.

Problem 3.22: The Intersection of a^x and $\log_a x$

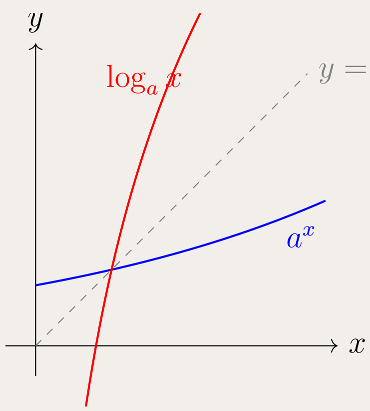
Investigate the number of real solutions to the equation $a^x = \log_a x$ for $a > 1$.

(i) Explain why the intersection points of these two inverse functions must lie on the line $y = x$.

(ii) Show that for $a = e$, there are no real solutions.

(iii) Find the value of a and the corresponding value of x for which the two curves are tangential to each other.

(iv) Determine the range of values for $a > 1$ such that the equation has exactly two distinct real solutions.



Hint: At the point of tangency on the line $y = x$, we must have $f(x) = x$ and $f'(x) = 1$. Use these two equations to solve for x and then find a.

Solution 3.22

(i) Reflection Property:
Since $g(x)$ is the inverse of $f(x)$, their graphs are reflections across $y = x$. For strictly increasing functions, if $a^x = x$ has a solution, then $\log_a(x) = x$ must share that same solution.
(ii) The Case for $a = e$:
The function $h(x) = e^x - x$ has a minimum at $x = 0$, where $h(0) = 1$. Since $e^x > x$ for all real x, the curves e^x and $\ln(x)$ never intersect.
(iii) The Critical Tangent:
We need $a^x = x$ and $\frac{d}{dx}(a^x) = 1$. $a^x \ln a = 1$. Since $a^x = x$, we have $x \ln a = 1 \implies a = e^{1/x}$. Substitute into $a^x = x$: $(e^{1/x})^x = x \implies e^1 = x$. Thus, $x = e$. The point of tangency is (e, e) and $a = e^{1/e}$.
(iv) Number of Solutions ($a > 1$):
If $a > e^{1/e}$: 0 solutions. If $a = e^{1/e}$: 1 solution. If $1 < a < e^{1/e}$: 2 solutions.

Takeaways 3.22

- **The Power of Symmetry:** You don't need to solve $\log_a(x) = a^x$. You only need to solve $a^x = x$.
- **Inverse Behavior:** If $f(x)$ is increasing, $f(x) \cap f^{-1}(x)$ occurs on $y = x$.

Problem 3.23: Uniform Convergence and the Limit–Integral Trap

In standard calculus, we often assume that "the limit of an integral is the integral of the limit." However, this is only guaranteed under **Uniform Convergence**. A sequence of functions $\{f_n\}$ defined on a set B converges uniformly to a function f if for all $\epsilon > 0$, there exists an N such that $|f(x) - f_n(x)| < \epsilon$ for all $n \geq N$ and all $x \in B$. If a sequence converges to f for each x individually but doesn't meet this "global" N requirement, it is called **pointwise convergence**.
Consider the sequence of functions

$$f_n(x) = n^2 x(1-x)^n, \qquad 0 \leq x \leq 1.$$

defined on the interval $x \in [0,1]$ for $n = 1, 2, 3, \ldots$.

(i) Show that for each fixed $x \in [0,1]$, we have $f_n(x) \to 0$ as $n \to \infty$.

(ii) Find the stationary point (the point where $f_n'(x) = 0$) of f_n on $[0,1]$ and hence determine its maximum value. Use this to show that $\{f_n\}$ does not converge uniformly to 0.

(iii) Evaluate

$$\int_0^1 \left(\lim_{n\to\infty} f_n(x)\right) dx \qquad \text{and} \qquad \lim_{n\to\infty} \int_0^1 f_n(x)\, dx.$$

Comment on the result.

(iv) Now consider $g_n(x) = \dfrac{x^n}{n}$ on $[0,1]$. Prove that $g_n \to 0$ uniformly.

Hint: For $0 < x < 1$, write $(1-x)^n = r^{-n}$ for some $r > 1$. In part (ii), differentiate and evaluate f_n at its critical point. In part (iii), use the substitution $u = 1 - x$. For part (iv), find the largest value of x^n/n on $[0,1]$.

Solution 3.23

(i) If $x = 0$ or $x = 1$, then $f_n(x) = 0$ for all n. If $0 < x < 1$, then $0 < 1 - x < 1$, so $(1-x)^n \to 0$ exponentially fast and therefore $n^2 x(1-x)^n \to 0$. Hence $f_n(x) \to 0$ pointwise on $[0, 1]$.
(ii)

$$f_n'(x) = n^2(1-x)^{n-1}\big(1 - (n+1)x\big).$$

So the stationary point is at $x = \dfrac{1}{n+1}$, which gives the maximum

$$M_n = f_n\left(\frac{1}{n+1}\right) = \frac{n^2}{n+1}\left(\frac{n}{n+1}\right)^n.$$

Since $\left(\dfrac{n}{n+1}\right)^n \to e^{-1}$, we get $M_n \sim \dfrac{n}{e} \to \infty$. In particular, $\max_{x\in[0,1]} |f_n(x) - 0| \le M_n$ does not tend to 0, so the convergence is not uniform.
(iii) Since the pointwise limit is 0,

$$\int_0^1 \left(\lim_{n\to\infty} f_n(x)\right) dx = \int_0^1 0\, dx = 0.$$

Also,

$$\int_0^1 f_n(x)\, dx = n^2 \int_0^1 x(1-x)^n\, dx.$$

Let $u = 1 - x$. Then

$$n^2 \int_0^1 (u^n - u^{n+1})\, du = n^2\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{n^2}{(n+1)(n+2)} \to 1.$$

Thus

$$\int_0^1 \left(\lim_{n\to\infty} f_n(x)\right) dx = 0 \qquad \text{but} \qquad \lim_{n\to\infty} \int_0^1 f_n(x)\, dx = 1.$$

So here taking limits and integrating cannot be interchanged.
(iv) For $0 \le x \le 1$,

$$0 \le g_n(x) = \frac{x^n}{n} \le \frac{1}{n},$$

with equality at $x = 1$. Hence

$$\max_{x\in[0,1]} |g_n(x) - 0| = \frac{1}{n} \to 0.$$

Therefore $g_n \to 0$ uniformly on $[0, 1]$.

Takeaways 3.23

- Pointwise convergence checks each fixed input separately, whereas uniform convergence controls the whole interval at once through the supremum error.
- A sequence can converge pointwise to 0 while still developing a narrow spike whose total area does not vanish.
- Uniform convergence is a safe condition for swapping limits and integrals; this example shows what can go wrong when uniform convergence fails.
- Uniform convergence links to differential equations and series expansions, where controlling the error across an entire domain is crucial for valid approximations.

4 Conclusion

Functions are one of the most unifying ideas in all of mathematics. Mastering the language of functions—domain, range, composition, and inverse—gives you the tools to read and write mathematics fluently across every topic in HSC Extension 1 and Extension 2. As this booklet grows, it will become a compact reference for the patterns and techniques that appear most often in HSC examinations.

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Repository: https://github.com/vuhung16au/math-olympiad-ml/tree/main/HSC-Functions

5 Appendix

5.1 Strict Definitions of Continuity

These strict definitions are very helpful for higher studies and for understanding mathematics, functions, and continuity in a rigorous way.

1. The limit definition (calculus). A function $f(x)$ is continuous at a point $x = c$ if and only if all three conditions hold:

- $f(c)$ exists.
- $\lim_{x\to c} f(x)$ exists (that is, the left-hand and right-hand limits both exist and are equal).
- $\lim_{x\to c} f(x) = f(c)$.

If even one condition fails at $x = c$ (for example, a hole, a jump, or a vertical asymptote), then f is discontinuous at c.

2. The ϵ-δ definition (real analysis). Let f be defined on a domain D, and let $c \in D$. We say f is continuous at c if:

> For every real number $\epsilon > 0$, there exists a real number $\delta > 0$ such that for all $x \in D$,
> $$|x - c| < \delta \implies |f(x) - f(c)| < \epsilon.$$

This is the most rigorous standard definition of continuity. It replaces informal language like "x approaches c" with precise inequalities.

5.2 Open and Closed Intervals

An interval is *closed* if it contains all of its endpoints, and *open* if it does not contain any of its endpoints.

Conceptual Examples:

a. Explain why the degenerate interval $[5, 5]$ is closed.
The interval $[5, 5]$ contains exactly one real number: 5. Its endpoints are 5 and 5. Since the interval contains 5, it contains all of its endpoints, meeting the exact definition of a closed interval.

b. Explain why the interval $(-\infty, \infty)$ is open.
An interval is open if it does not contain any endpoints. The interval $(-\infty, \infty)$, representing the entire real line $\mathbb{R}$, has no boundaries or endpoints in the real numbers. Because there are no endpoints to contain, it vacuously satisfies the condition of not containing any of its endpoints.

c. Explain why the interval $(-\infty, \infty)$ is closed.
An interval is closed if it contains *all* of its endpoints. Since it has zero endpoints, there are no endpoints missing from the set. Therefore, it vacuously contains all of its endpoints. In topology, spaces with this property are considered both open and closed (clopen).

5.3 Table of Variation

Outside the NESA syllabus. The *Table of Variation* (also called a *variation table*) is **not part of the NSW HSC Mathematics Extension 1 or Extension 2 syllabuses**. For HSC work, stay with the simpler **sign tables** you already use to locate zeroes, test intervals, and decide whether a stationary point is a maximum or minimum.

At university level, however, the Table of Variation is a standard tool: a structured table (or grid) that records the full global behaviour of a differentiable function *before* you sketch its graph. Where a high-school sign table might only track the sign of an expression, a variation table also records critical values, limits at boundaries and asymptotes, and the direction of increase or decrease on every interval.

Typical rows.

- **Domain (x):** from $-\infty$ to $+\infty$, listing stationary points ($f'(x) = 0$), points where f' is undefined, and vertical asymptotes (often marked with a double line $\|$ through the table).
- **First derivative ($f'(x)$):** signs $+$, 0, or $-$ on each interval.
- **Variation ($f(x)$):** arrows $\nearrow$ (increasing) and $\searrow$ (decreasing), with exact values $f(c)$ at extrema and evaluated limits at $\pm\infty$ and near asymptotes.

A fourth row for $f''(x)$ (concavity and inflection) appears in more advanced courses; it is optional here.

Once the table is complete, sketching the graph is largely a matter of following the arrows and plotting the key coordinates.

Example (a). $y = x^4 + 2x^2 - 2$.
Differentiate:

$$f'(x) = 4x^3 + 4x = 4x(x^2 + 1).$$

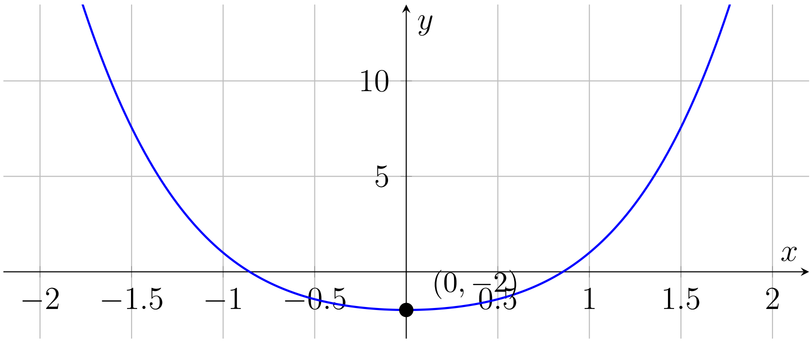
Since $x^2 + 1 > 0$ for all real x, we have $f'(x) = 0$ only at $x = 0$, and

$$f'(x) < 0 \text{ for } x < 0, \qquad f'(x) > 0 \text{ for } x > 0.$$

Also $f(0) = -2$, and as $x \to \pm\infty$, the x^4 term dominates so $f(x) \to +\infty$. The graph is even (only even powers of x).

x	$-\infty$	$(-\infty, 0)$	0	$(0, \infty)$	$+\infty$
$f'(x)$		$-$	0	$+$	
$f(x)$	$+\infty$	$\searrow$	-2	$\nearrow$	$+\infty$

Sketch. A symmetric "U-shaped" curve with a single minimum at $(0, -2)$, falling from $+\infty$ as $x \to -\infty$ and rising back to $+\infty$ as $x \to +\infty$.



Example (b). $y = \dfrac{x+1}{x-1}$.

Domain: $x \neq 1$. Rewrite as $y = 1 + \dfrac{2}{x-1}$ to read off the horizontal asymptote $y = 1$ and vertical asymptote $x = 1$. The zero is $x = -1$. Differentiate:

$$f'(x) = \frac{(x-1)-(x+1)}{(x-1)^2} = \frac{-2}{(x-1)^2} < 0$$

for all x in the domain, so f is strictly decreasing on $(-\infty, 1)$ and on $(1, \infty)$—there are no turning points.

Limits:

$$\lim_{x\to-\infty} f(x) = 1^-, \quad \lim_{x\to 1^-} f(x) = -\infty, \quad \lim_{x\to 1^+} f(x) = +\infty, \quad \lim_{x\to+\infty} f(x) = 1^+.$$

Key values: $f(-1) = 0$ and $f(0) = -1$.

x	$-\infty$	$(-\infty,-1)$	-1	$(-1,1)$	$\Vert$	1	$\Vert$	$(1,\infty)$	$+\infty$
$f'(x)$		$-$		$-$				$-$	
$f(x)$	1^-	$\searrow$	0	$\searrow$		$-\infty$ / $+\infty$		$\searrow$	1^+

Sketch. Two decreasing branches separated by the vertical asymptote $x = 1$. The left branch passes through $(-1, 0)$ and $(0, -1)$, approaches $y = 1$ from below as $x \to -\infty$, and drops to $-\infty$ as $x \to 1^-$. The right branch comes down from $+\infty$ as $x \to 1^+$ and approaches $y = 1$ from above as $x \to +\infty$.